* Errors in Gear Measurement

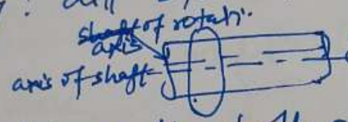
1. Profile error: Max^m distance of any point on the tooth profile from and normal to the design profile.

2. Cyclic error: it occur during each revolution of the gear.
6 x
breakage of tooth.

3. Periodic error: error which occurs at regular intervals.

4. Runout: total range of reading of a fixed indicator at a point to a rotating surface.
   - Radial runout: measured along ⊥ to the axis of rotation
   - Axial " : " // " "

5. Eccentricity: diff. b/w axis of rotation & geometrical axis



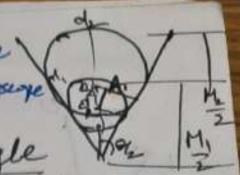
gear rotates some out of axis.

6. Backlash: b/w mating tooth profile
   = tooth space — tooth thickness

Parkinson Gear Tester

* Screw thread Measurement *

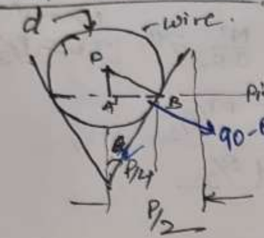
Pitch measuring m/c / Tool maker's microscope

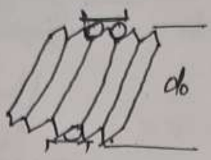


① Pitch P —— by thread gauge
screw pitch gauge

② ~~Flank angle~~

② Best wire size





* Flank angle

$$\sin\theta/2 = \frac{O_2A}{O_1O_2}$$

$$= \frac{(d_2-d_1)/2}{\left(\frac{M_2}{2}-\frac{d_2}{2}\right)-\left(\frac{M_1}{2}-\frac{d_1}{2}\right)}$$

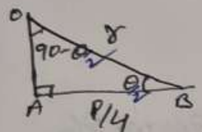
$$\boxed{\sin\theta/2 = \frac{d_2-d_1}{(M_2-M_1)-(d_2-d_1)}}$$

for metric thread $\theta = 60°$

$AB = P/4$

$\angle OBA = 90-(90-\frac{\theta}{2}) = \theta/2$

$\angle AOB = 90-\theta/2$



$\sin(90-\frac{\theta}{2}) = \frac{AB}{OB} \Rightarrow \cos\frac{\theta}{2} = \frac{P/4}{r}$

$r = \frac{P}{4}\sec\frac{\theta}{2} \Rightarrow \boxed{d = \frac{P}{2}\sec\frac{\theta}{2}}$

$\boxed{d = 0.5774P}$

③ Minor diameter — by point micrometer

for Int. 1. taper parallels 2. Rollers with slip gauge

$d_1 = D \pm (M_2 - M_1)$

D = length slip gauge + 2 × dia of Roller.

④ Major diameter (Nominal diameter) – by floating carriage Micrometer

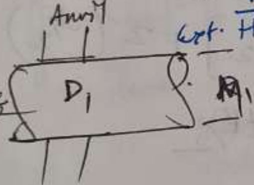
Bench Micrometer

Micrometer

| for Int. thread. Replica using POP / Silica

Two wooden box over std. Cyl

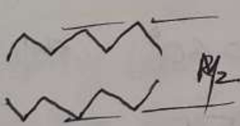
Ext. thread –



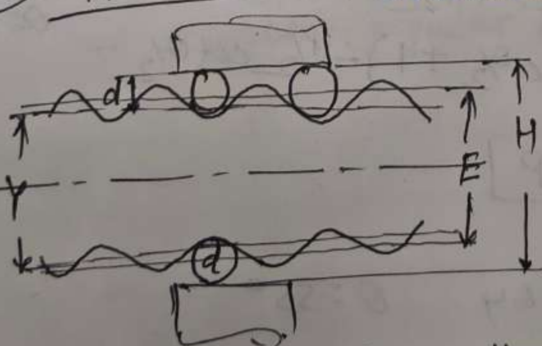
$$\boxed{d_0 = D_1 \pm (M_2 - M_1)}$$

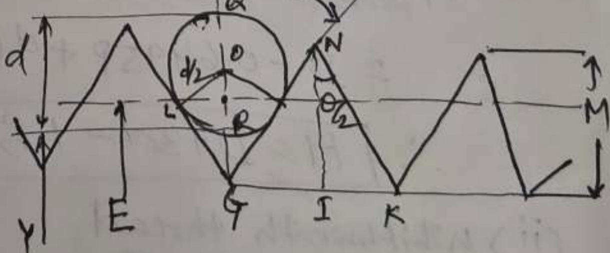
D_1 – dia of std. Cyl.

$(M_2 - M_1)$ – Micrometer Reading over screw thread



⑤ effective dia. (Three wire method)
Screw thread Micrometer





E – effective dia of screw thread / pitch dia

H – diameter over wire (Major)

Y – dia under wire (Min) $[H-2d]$

d – dia of wire.

Gear Parameters Measurement using Gear Tooth Vernier Caliper

① Outside dia (OD) — MSR + VSR + LC
$60 + 0 \times 0.02 = 60mm$

② Gear tooth thickness — MSR $4 + 10 \times 0.02 = 4.20mm$

③ Depth of gear tooth — $3 + 35 \times 0.02 = 3.70mm$

④ PCD (Pitch Circle dia)
Imaginary Circle — where tooth touches to another gear's teeth

T = No. of teeth = 28

$D = T \times OD / T+2$
$= 28 \times 60 / 28+2 = 56$ mm.

⑤ Module (M) — length of PCD per tooth

$m = \frac{D}{T} = \frac{56}{28} = 2$ mm

⑥ Circular pitch: Arc distance measured around the pitch circle from flank of one tooth to the flank of next tooth [same side of next tooth].

$CP = \frac{\pi D}{T} = \pi m = 3.14 \times 2 = 6.28$ mm

⑦ Addendum: Radial distance, measured from top land of tooth to the pitch circle.

Add. $= 1m = 2$ mm

⑧ Clearance: Radial distance from tip of tooth to the bottom of mating tooth space when teeth are symmetrically engaged.

std. values $= 0.157m$ or $0.25m$
$= 0.314$ m 0.50 mm

⑨ Dedendum: Radial distance measured from PCD to the bottom land of gear tooth

Ded. = Add. + Clearance
$= 2 + 0.314 = 2.314$ mm
or
$= 2 + 0.50 = 2.50$ mm.

(10) Chordal addendum — set on vertical scale with help of slip gauges

$$d = \left(\frac{Tm}{2}\right)\left[1+\left(\frac{2}{T}\right) - \cos\left(\frac{90}{T}\right)\right]$$

$$= \left(\frac{28\times 2}{2}\right)\left[1+\left(\frac{2}{28}\right) - \cos\left(\frac{90}{28}\right)\right]$$

$= 2.044$ mm by slip gauges.

1.004 ✓
1.04 ✓

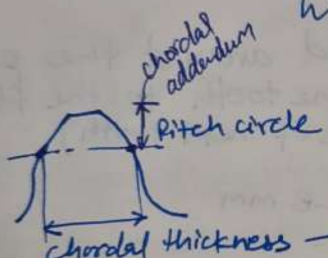
Select tooth
Chordal thickness $W = MSR + VSR \times LC$
$= 3 + 2 \times 0.02 = 3.04$ mm

· 3 Readings, then average.

Theoretical formula for chordal thickness

$$W = Tm\left[\sin\left(90/T\right)\right]$$

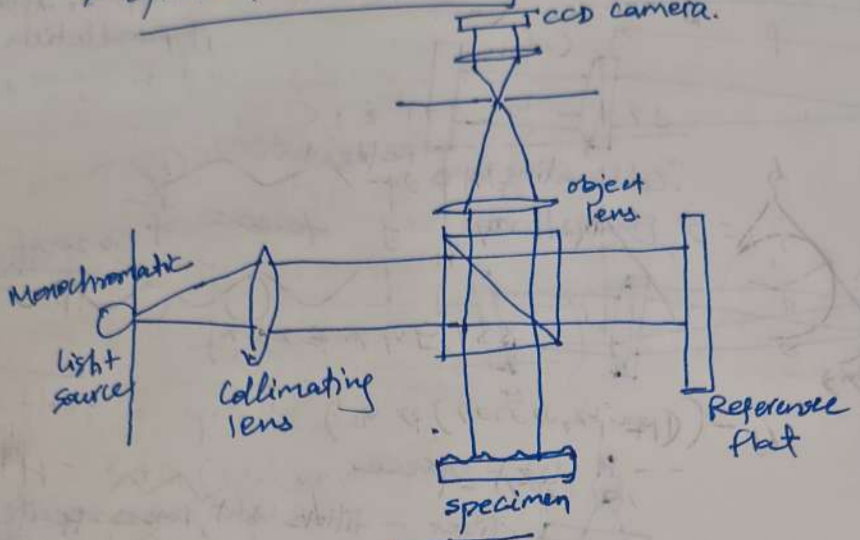
$$= 28\times 2\left[\sin\left(90/28\right)\right]$$

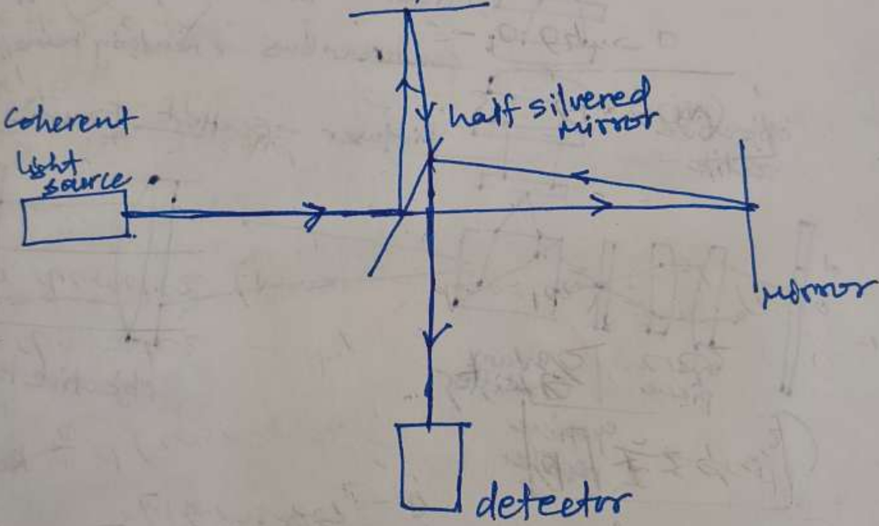


→ Tooth thickness measured along a chord passing through the points where the pitch circle crosses the tooth profile.

(11) Runout : eccentricity in the pitch circle
└ give periodic vibration during each revolution of the gear
└ give the tooth failures in gears.
Runout is measured by means of eccentricity testers.

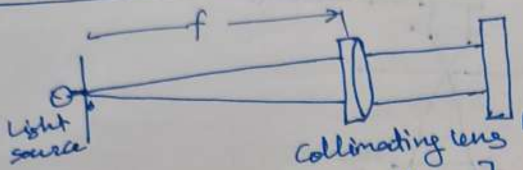
* Optical Interferometry — Study surface structure





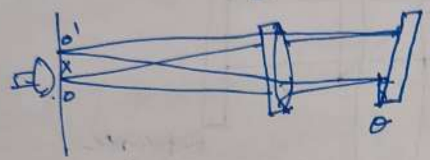
Auto collimator

— Measuring Angle, straightness, flatness, squareness + parallelism.

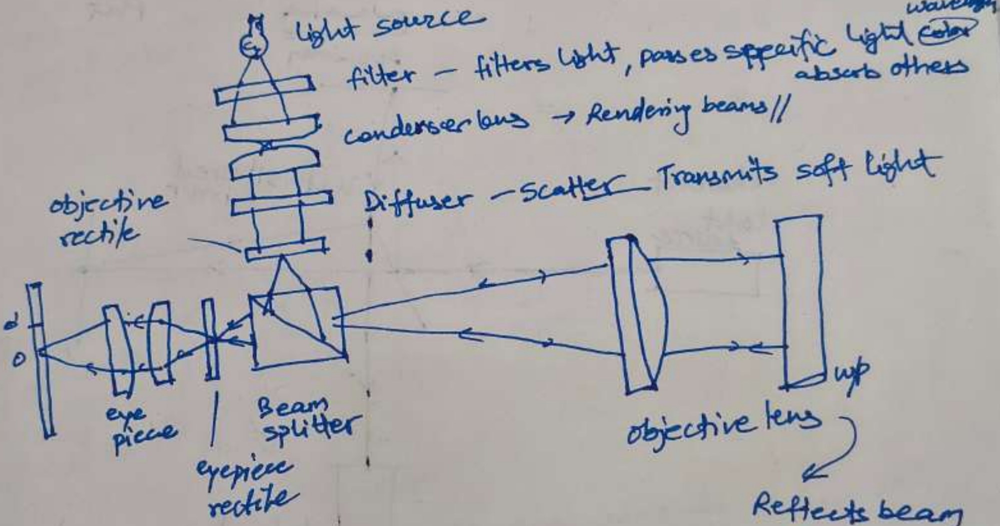


[Makes // rays]

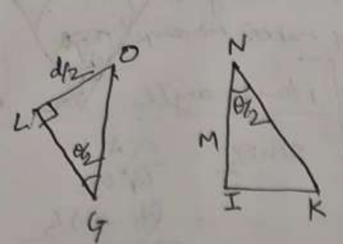
$\tan 2\theta = \frac{x}{f} \Rightarrow 2\theta \times f = x$



$x = 2\theta \times f$



Appl^n - Aligning laser rod ends
- Checking face parallelism of optical windows + wedges
- ~~Ede~~ Monitoring angular movement over long periods of time
- checking angular position in mech-system



$\Delta OLG \Rightarrow \sin(\theta/2) = \frac{OL}{OG}$

$\frac{d}{2}\operatorname{cosec}(\theta/2) = OG$

$\Delta NIK \quad \tan\theta/2 = \frac{IK}{NI}$

$\cot\theta/2 = \frac{NI}{IK} \Rightarrow \quad NI = M, \quad IK = P/2$

$\cot\theta/2 = M/P/2$

$\Rightarrow M = P/2 \cot\theta/2$

$\therefore RG = \frac{1}{2} \times M = \frac{1}{2} \times P/2 \cot\theta/2$

$$\boxed{RG = P/4 \cot\theta/2}$$

$OR + RG = OG \Rightarrow OR = OG - RG$

$= \frac{d}{2}\operatorname{cosec}\theta/2 - P/4 \cot\theta/2$

$H = E + 2(OR) + 2(OQ)$

$= E + 2\left[\frac{d}{2}\operatorname{cosec}\theta/2 - P/4 \cot\theta/2\right] + 2(d/2)$

$\therefore H = E + d(\operatorname{cosec}\theta/2 + 1) - P/2 \cot\theta/2$

(i) for Metric thread. Std. Nomencl.

depth of thread $= 0.6495$, $\theta = 60°$

$\therefore$ effective dia $= E = D - 0.6495P$ (D: Major Dia)

$\therefore H = E + d(\operatorname{cosec}\theta/2 + 1) - P/2 \cot\theta/2 \qquad \theta = 60°$

$= D - 0.6495P + d(\operatorname{cosec}\theta/2 + 1) - P/2 \cot\theta/2$

$$\therefore \boxed{H = D + 3d - 1.5155P}$$

(ii) Whithworth thread

depth of thread $= 0.64 \qquad \theta = 55°$

$E = D - 0.64P$

$$\therefore \boxed{H = D + 3.1656d - 1.6P}$$

...Quality Management (TQM) is a management framework focused on continuous improvement, customer satisfaction, and total employee involvement to enhance long-term success. It integrates quality principles into all organizational processes—products, services, and culture—to improve efficiency and meet or exceed customer expectations.

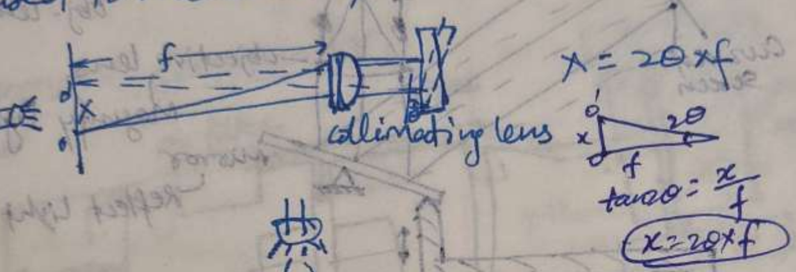
Key Principles of TQM

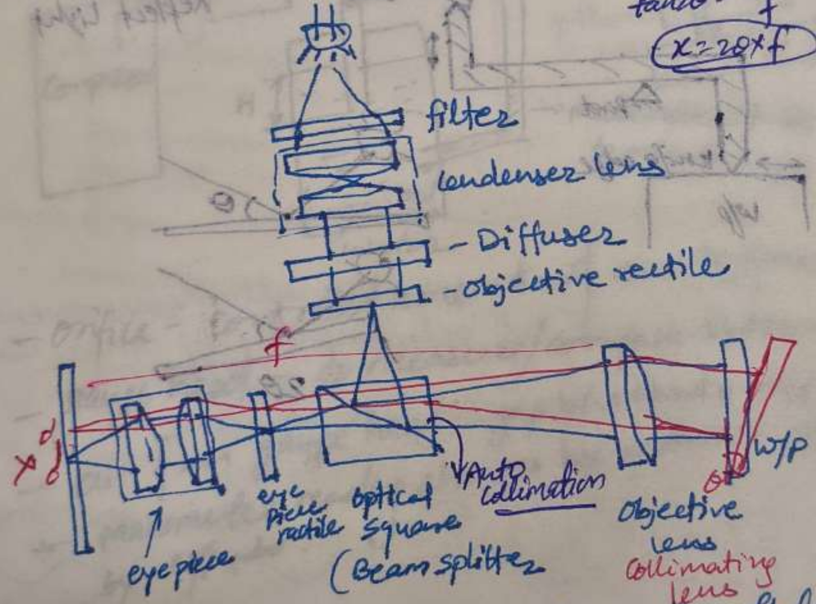
TQM is built on several core pri... goals:

Autocollimator

- Visual — 4.85 micro radians
- Digital — 100 times more resolution

- Measure small angular diff.
- Used to align optical components
- Used to measure optical & Mech. deflection



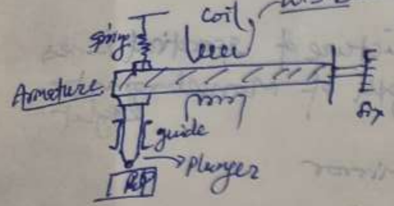


Diffuser — Scatter light rays to transmit soft lights

Appⁿ.
- Checking straightness of machine tool sideways
- Checking parallelism
- Checking squareness of column to base
- Checking small linear displacements
- Measuring very small angles.

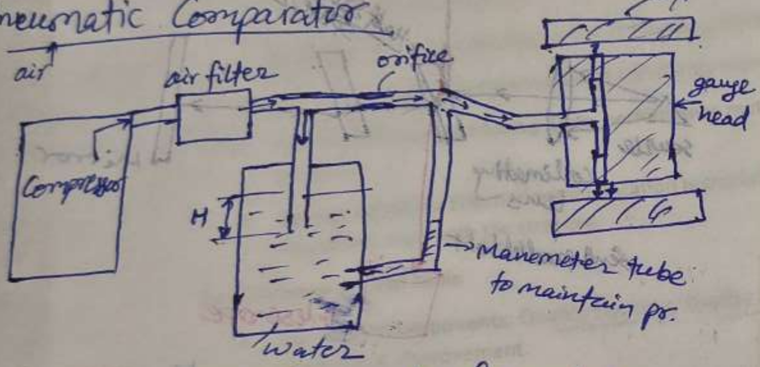
Total Quality Management (TQM) is a management framework focused on continuous improvement. It integrates ... total employee involvement to enhance long-term success ... products, services, and culture—to improve

① Electrical Comparator



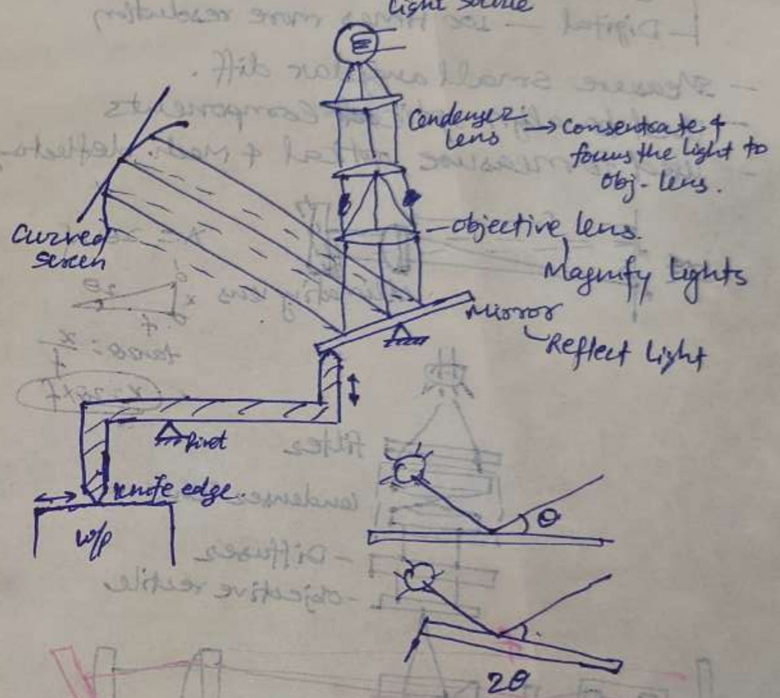
① set at zero
↓
Balance
WSB → Unbalance
→ Measure deviation

② Pneumatic Comparator



- Orifice - controlled amount of air to pass
- gauge head - to measure/compare Internal dia
- air from gauge head - gap b/w head & w/p.
+ Manometer reading changes by restriction of passage by w/p.

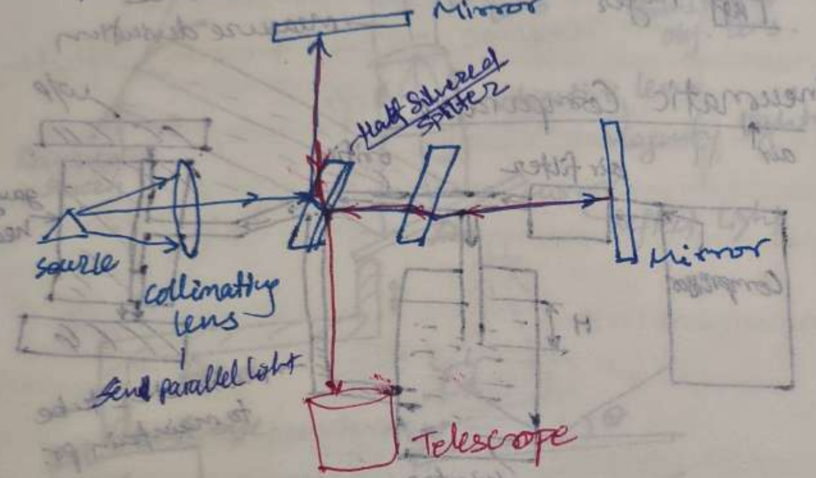
Mechanical—
④ Optical Comparator



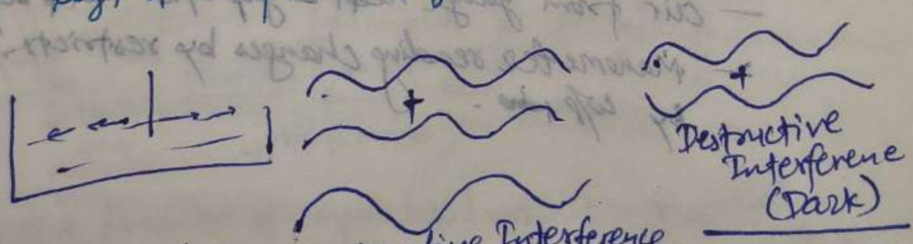
TQM Total Quality Management (TQM) is a management framework focused on continuous improvement, customer satisfaction, and total employee involvement ... quality principles into all ... It integrates ... improve

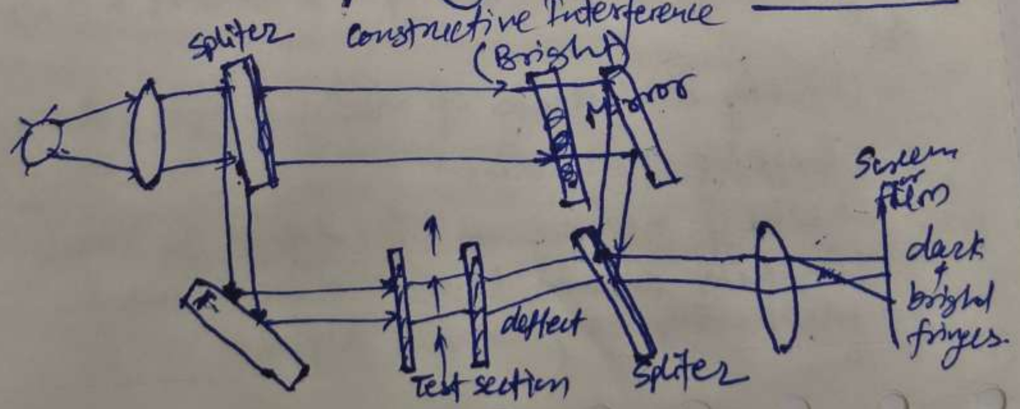
Optical Interferometer

- Used to study fine structure of spectral lines
- To determine wavelength of Monochromatic light
- flow visualization



- Used to make very small measurements that are not achievable any other way
- eg: length, surface irregularities





SQC

→ Implementation of quality at policy, design, mfg & installation stages is essential to ensure that the end product or service meets the required quality standards. Some steps that can be taken to implement quality at each stage are as follows:

① Policy stage:

At this stage, quality can be implemented by setting policies and procedures that promote quality throughout the organization. Following are the steps:

→ Develop a quality policy statement that outlines the company's commitment to quality.

→ Establish quality objectives that align with the company's overall goals.

→ Define procedures for implementing and monitoring quality standards.

→ Assign roles & responsibilities to ensure that quality standards are met.

② Design stage:

→ At this stage, quality can be implemented by ensuring that the design meets the required specifications & standards. Following steps can be taken:-

Key Aspects of the ISO 9000 Family:

- **ISO 9000:2015:** Covers fundamentals and terminology for QMS.
- ISO 9001:2015: Sets the requirements for a QMS, which organizations can be certified against.
- **Seven Quality Management Principles:**
  1. Customer focus
  2. Leadership
  3. Engagement of people
  4. Process approach
  5. Improvement
  6. Evidence-based decision making
  7. Relationship management

Benefits of ISO 9000:

- **Enhanced Customer Satisfaction:** Ensures consistent quality.
- **Improved Efficiency:** Reduces waste and increases operational efficiency.
- **Global Recognition:** Facilitates trade by meeting internationally recognized standards.
- **Continuous Improvement:** Promotes a culture of ongoing refinement.

Industry Application:

The standards are designed to be general and applicable to any organization, regardless of size or industry. Specific industry sectors, such as automotive, aerospace, and medical devices, often have specialized versions based on the core ISO 9001 standard.

Certification Process:

Organizations often use third-party auditors to certify their compliance with the standard. The ISO 9000:2015 version is currently the standard for Fundamentals and Vocabulary.

TQM Total Quality Management (TQM) is a management framework focused on continuous improvement, customer satisfaction, and total employee involvement to enhance long-term success. It integrates quality principles into all organizational processes—products, services, and culture—to improve efficiency and meet or exceed customer expectations.

Key Principles of TQM

TQM is built on several core principles focused on long-term sustainability rather than short-term goals:

- **Customer-Focused:** The customer ultimately determines the level of quality.
- **Total Employee Involvement:** All employees work toward common goals, empowering them to take ownership of quality.
- **Process-Centered:** Focuses on analyzing, improving, and controlling internal processes to achieve quality.
- **Integrated System:** While processes vary, TQM acts as an overarching framework that connects business functions.
- **Strategic and Systematic Approach:** A strategic plan is adopted to achieve quality goals.
- **Continual Improvement:** Emphasizes constantly improving operational efficiency and quality.
- **Fact-Based Decision Making:** Data, performance metrics, and tools are used to measure progress.
- **Communications:** Effective communication is crucial to maintain motivation and ensure all employees align with the strategy.

Key Components and Tools

- **Four Main Components:** Quality Planning, Quality Assurance, Quality Control, and Continuous Improvement.
- **Tools for Improvement:** Often involves using data-driven tools such as check sheets, control charts, Pareto charts, histograms, and cause-and-effect diagrams.

Benefits of TQM

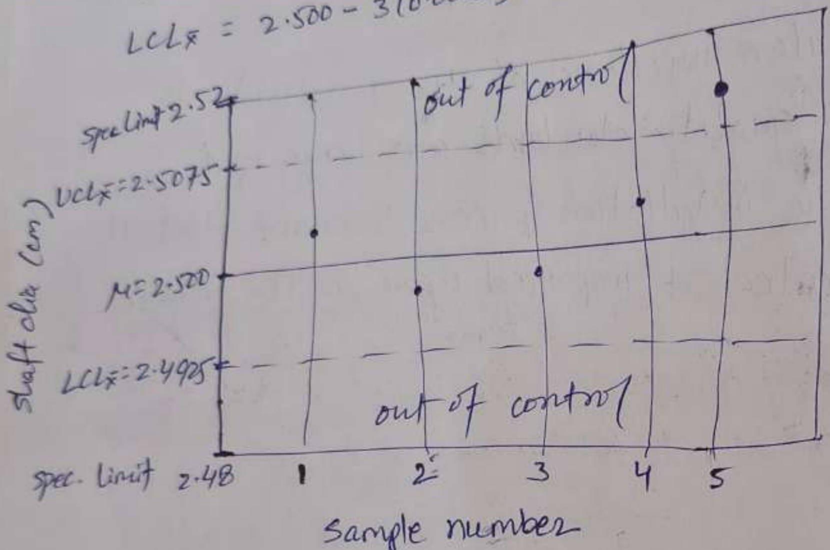
- **Improved Quality:** Reduction in errors and defects, leading to higher product and service quality.
- **Enhanced Customer Satisfaction:** By delivering better quality, customer satisfaction increases.
- **Reduced Costs:** Lower costs associated with rework, defects, and waste.
- **Improved Employee Engagement:** Empowerment and training improve employee morale and skills.

ISO 9000 The ISO 9000 family is a set of international standards for Quality Management Systems (QMS), developed by the International Organization for Standardization (ISO). It helps organizations ensure products and services consistently meet customer requirements, focusing on quality principles, continuous improvement, and customer satisfaction.

metal shaft of dia. ~~2.5 2.500 ± 0.020~~ 2.500 ± 0.020 cm. STD, σ estimated approx → 0.005. If sample size is 4, then $\sigma_{\bar{x}} = 0.005/\sqrt{4} = 0.00$

$UCL_{\bar{x}} = 2.500 + 3(0.0025) = 25075$

$LCL_{\bar{x}} = 2.500 - 3(0.0025) = 2.4925$



$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ | $UCL = x + 3\sigma_{\bar{x}}$
$LCL = x - 3\sigma_{\bar{x}}$

SD
Ab 03
47

Develop an installation plan that ~~includes~~ includes procedures for ensuring that the installation is done correctly

→ Train personnel in the proper installation procedures & techniques.

→ Conduct regular inspections of the installation to ensure that quality standards are being met.

→ Document the installation process to ensure that it can be replicated & improved upon in the future.

→ Clearly define design requirements + specifications.
→ Conduct design reviews to ensure that the design meets the required standards.
→ Use appropriate design tools + techniques to ensure that the design is robust + reliable.
→ Conduct testing + validation to ensure that the design meets the required quality standards.

③ Manufacturing stage:

At this stage, quality can be implemented by ensuring that the manufacturing process is controlled + monitored to meet the required quality standards. Following steps can be taken:

→ Develop a mfg plan that includes procedures for controlling + monitoring the process.
→ Implement process controls such as SPC (statistical process control) to monitor process variability + take corrective actions when necessary.
→ Conduct regular audits of the mfg process to ensure that quality standards are being met.
→ Train personnel in the proper use of equipment + procedures to ensure that quality standards are being met.

④ Installation stage:

At this stage, quality can be implemented by ensuring that the installation is done correctly + meets the required quality standards. Following steps can be taken:-

Calculations :-

↳ no. of samples = 5 (sample size)

→ $n = 20$

→ $A_2 = 0.58$

Factor for $\bar{R}$ chart , $D_3 = 0$, $D_4 = 2.11$

→ Control line ,

$$\bar{\bar{X}} = \frac{185.94}{20} = 9.297 \approx 9.3.$$

$$\bar{R} = \frac{\Sigma R}{n} = \frac{2.17}{20} = 0.108 \approx 0.11$$

→ For $\bar{X}$ chart ,

$$U.C.L. = \bar{X} = \bar{\bar{X}} + A_2\bar{R}$$

$$= 9.3 + 0.58 \times 0.11$$

$$= 9.363$$

$$L.C.L. = \bar{X} = \bar{\bar{X}} - A_2\bar{R}$$

$$= 9.3 - 0.58 \times 0.11$$

$$= 9.236$$

→ For $\bar{R}$ Chart ,

Centre line , $\bar{R} = 0.11$

$$U.C.L_R = D_4\bar{R} = 2.11 \times 0.11 = 0.232$$

$$L.C.L_R = D_3\bar{R} = 0 \times 0.11 = 0$$

the frequency distribution process

Procedure :-

a) Select the appropriate quality characteristics to be studied

b) Record data : required no of samples compared to an adequate no. of parts and size

c) Determine the control limit from the data

d) Check if these control limits are economically satisfactory for the job.

e) Plot the control limit on suitable graph paper

f) Start the plotting of recorded result of production sample within or exceeding the control limit.

Formulae used :-

→ For $\bar{X}$ Chart :

When range is used as a measure of speed,

(i) Centre Line - $\bar{\bar{X}} = \Sigma\bar{X}/n$

(ii) U.C.L. (upper control limit) - $\bar{\bar{X}} + A_2\bar{R}$

(iii) L.C.L. (lower) - $\bar{\bar{X}} - A_2\bar{R}$

where,

n = total sample

A_2 = constant

$\bar{R} = \Sigma R/n$

estimated
$05/\sqrt{4} = 0.0025$

EXPERIMENT No. - 3

Object :- (A) To prepare the $\bar{X}$ and $\bar{R}$ chart for the given sample of product

(B) To show that distribution of sample mean from any universe follow normal distribution. Further analyse the effect of sample size on distribution

Theory :-
A Control Chart can be defined as a chronological graphical comparison of actual product quality characteristics with the information gathered during inspection of products
Types of Control Chart :-
(i) $\bar{X}$ and $\bar{R}$ Chart
(ii) P chart (based on fraction defectives)

$\bar{X}$ and $\bar{R}$ Chart - These charts are known as Control Chart for variables. Since these charts give the control limit for measuring the variation in quality characteristics. The basic principle of measurement of control chart limit is similar to that of completing

→ For $\bar{R}$ Chart : (K = sub group)

(i) Centre line - $\bar{R} = \Sigma R/K$

(ii) U.C.L.(R) - $D_4 \cdot \bar{R}$

(iii) L.C.L.(R) - $D_3 \cdot \bar{R}$

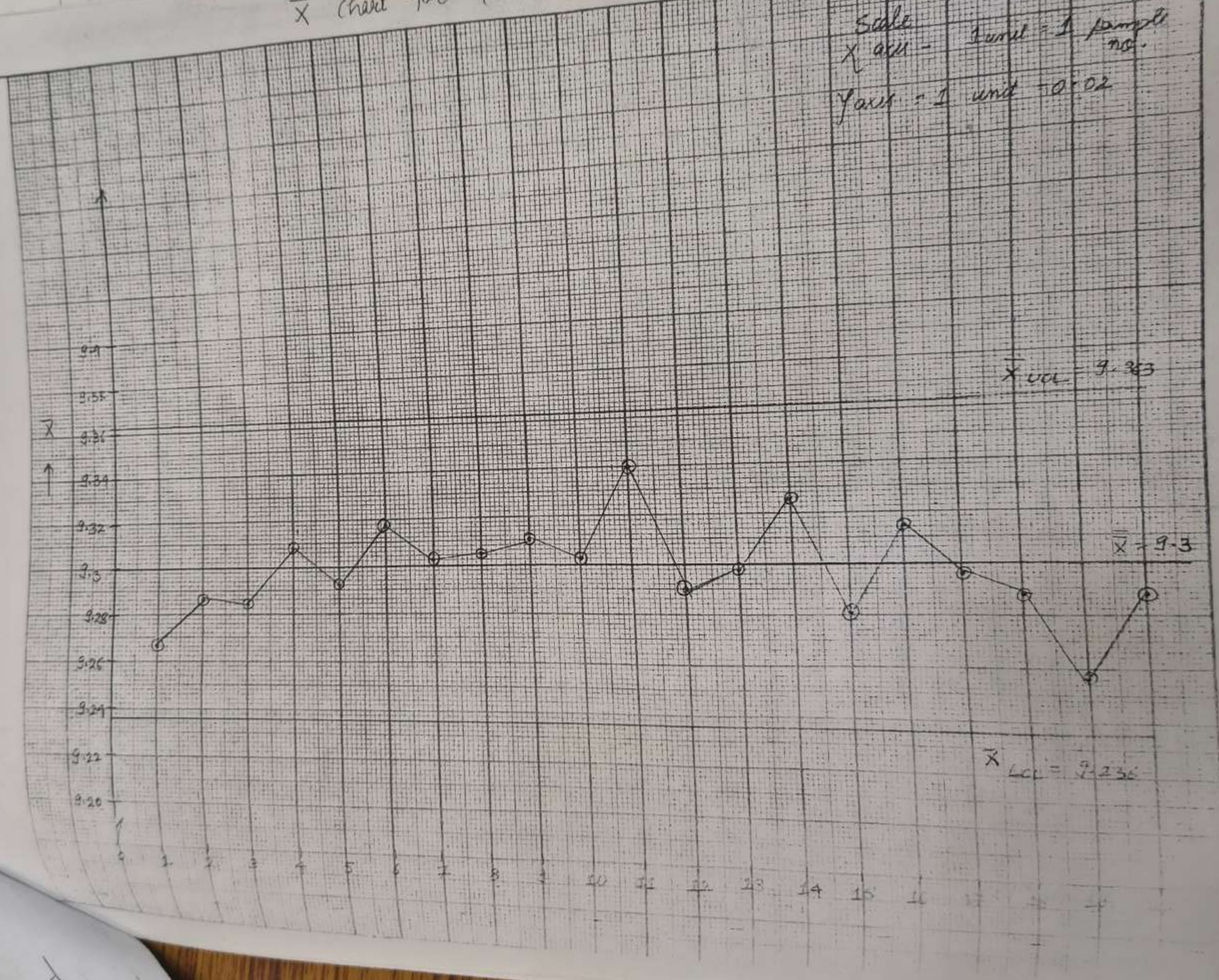
where,

D_3 and D_4 are constants that depends upon the sample size from which that particular sample is obtained.

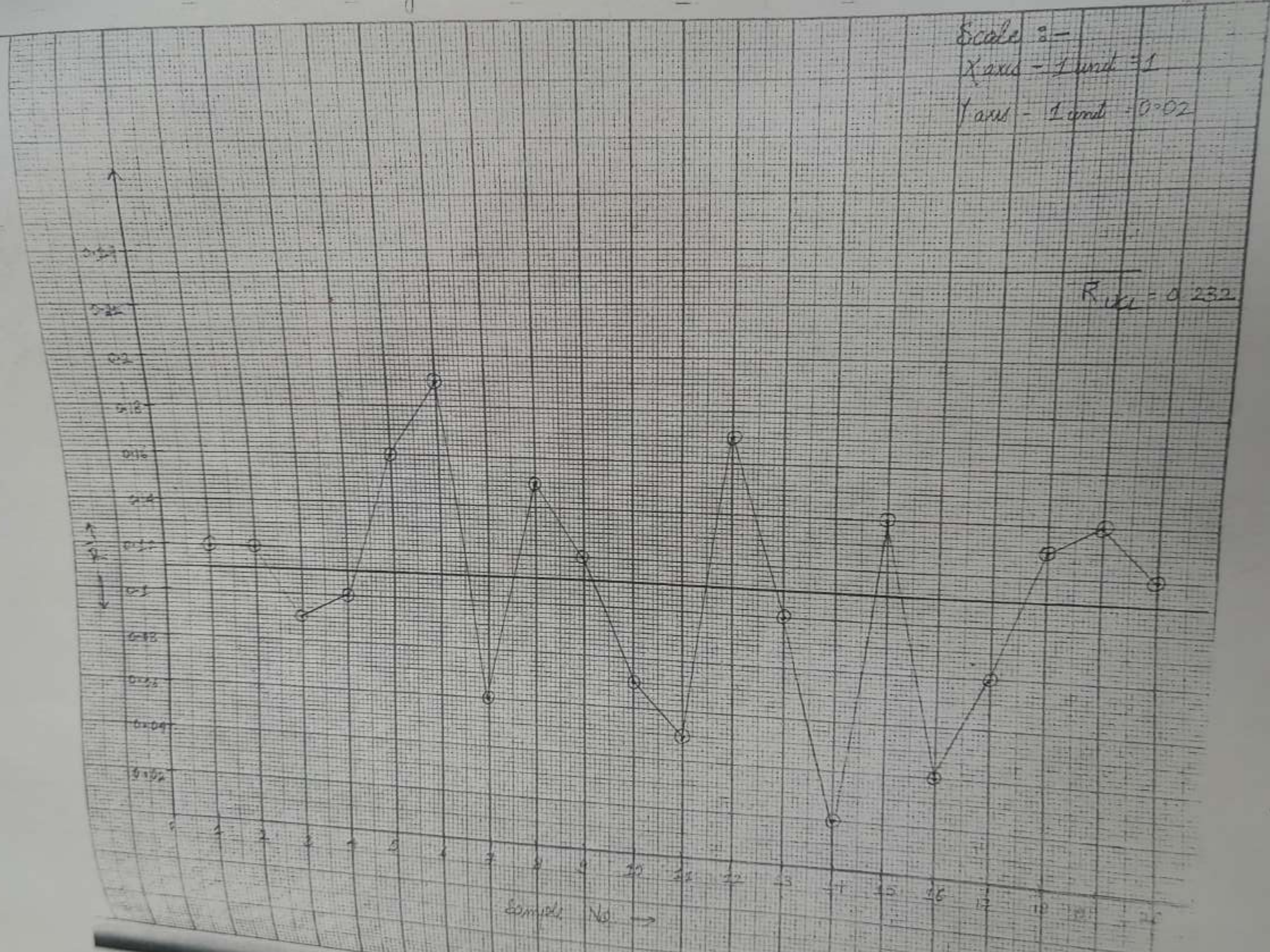
Observations :- A) Sample Size = 5

S.No	Sample Elements					Average	Range
	X_1	X_2	X_3	X_4	X_5	$\bar{X} = \Sigma X/n$	$(X_{max} - X_{min})$
1	9.18	9.28	9.28	9.29	9.3	9.266	0.12
2	9.29	9.32	9.31	9.2	9.31	9.286	0.12
3	9.32	9.31	9.24	9.32	9.23	9.284	0.09
4	9.25	9.32	9.35	9.31	9.31	9.308	0.1
5	9.33	9.33	9.31	9.32	9.17	9.292	0.16
6	9.37	9.38	9.33	9.19	9.32	9.318	0.19
7	9.27	9.32	9.27	9.32	9.33	9.302	0.06
8	9.33	9.19	9.34	9.34	9.32	9.304	0.15
9	9.35	9.33	9.23	9.32	9.32	9.31	0.12
10	9.33	9.26	9.28	9.32	9.32	9.302	0.07
11	9.31	9.35	9.36	9.36	9.32	9.34	0.05

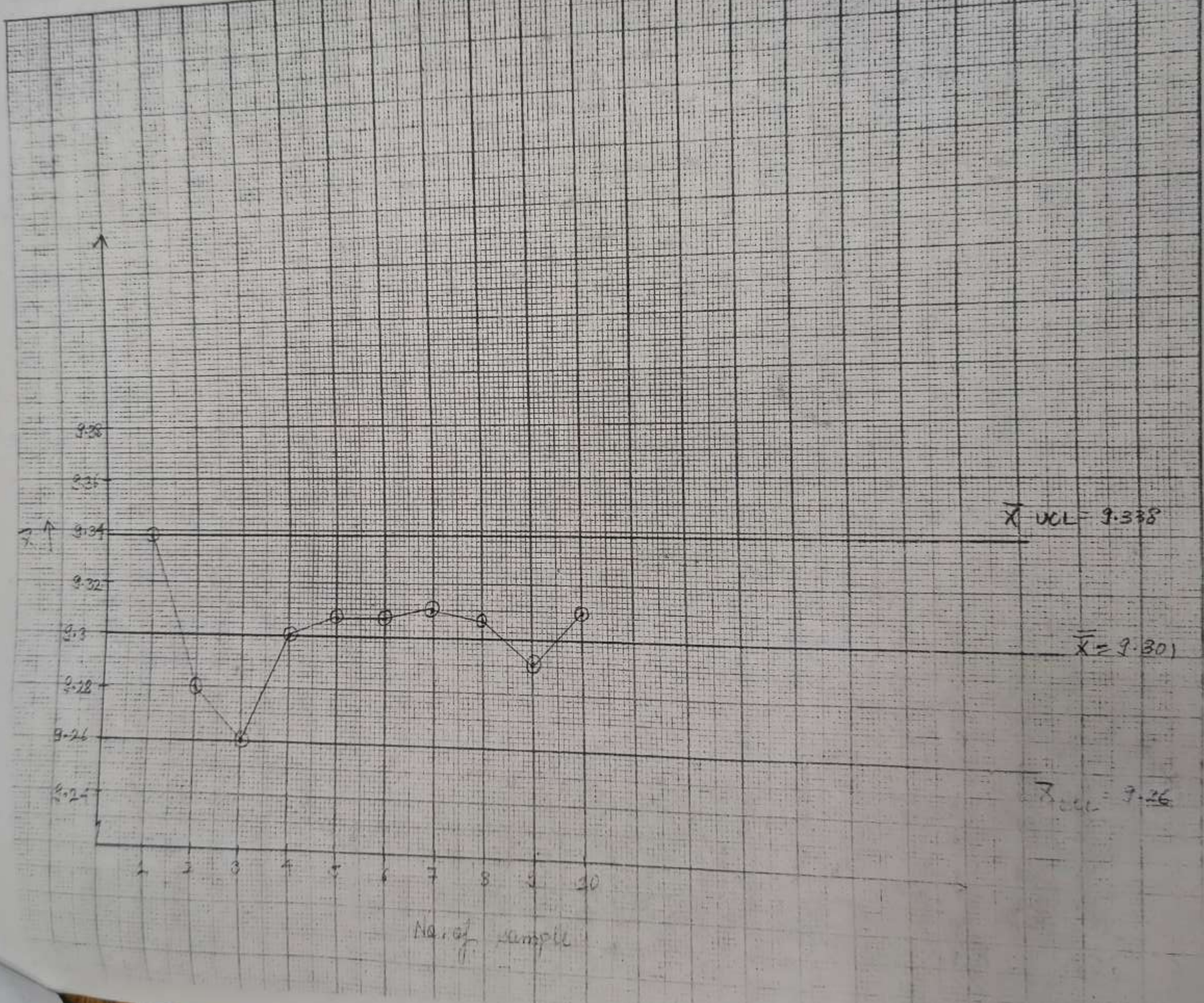
$\bar{X}$ chart for (N=20)



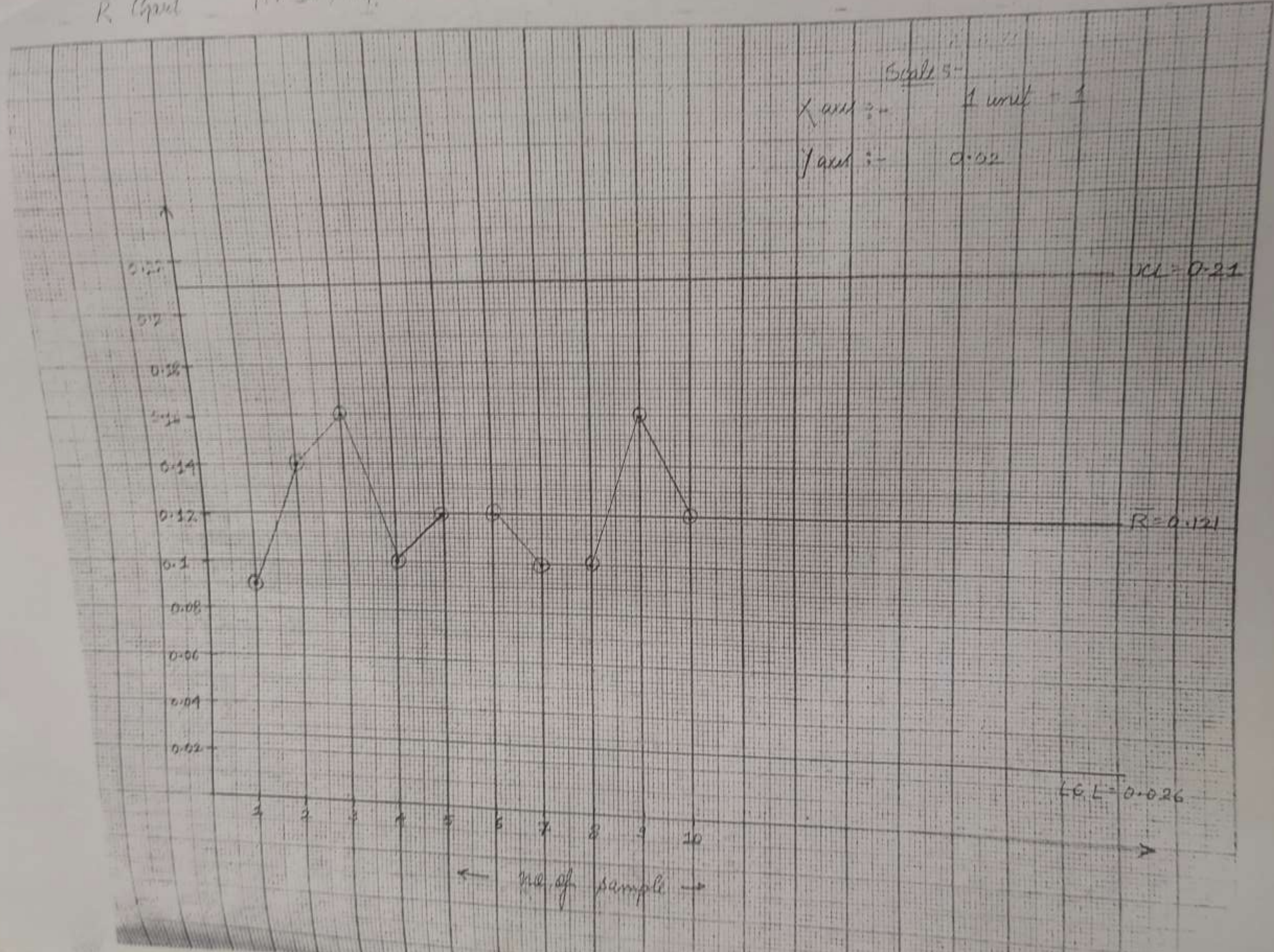
$\bar{R}$ chart for (N=20)



$\bar{X}$ Chart for N=10



$\bar{R}$ Chart (1-10 sample)



S.No.	Sample Elements					Average	Range
	x_1	x_2	x_3	x_4	x_5	$\bar{x} = \Sigma x/n$	$(x_{max} - x_{min})$
12	9.17	9.26	9.34	9.33	9.34	9.288	0.17
13	9.28	9.33	9.23	9.32	9.33	9.298	0.1
14	9.32	9.34	9.33	9.32	9.32	9.326	0.02
15	9.18	9.27	9.32	9.31	9.32	9.28	0.14
16	9.31	9.32	9.33	9.32	9.29	9.314	0.04
17	9.25	9.33	9.33	9.32	9.25	9.296	0.08
18	9.34	9.23	9.33	9.34	9.2	9.288	0.13
19	9.31	9.23	9.33	9.19	9.22	9.256	0.14
20	9.26	9.31	9.21	9.33	9.33	9.288	0.12
						$\Sigma \bar{x} = 185.94$	$\Sigma R = 2.17$

(B) Sample size - 10

S.No.	Sample elements										Average	Range
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	$\bar{x} = \Sigma x/n$	$x_{max} - x_{min}$
1)	9.24	9.33	9.33	9.32	9.33	9.25	9.32	9.32	9.33	9.33	9.338	0.09
2)	9.34	9.2	9.33	9.22	9.24	9.23	9.33	9.33	9.33	9.32	9.28	0.14
3)	9.32	9.27	9.19	9.23	9.2	9.33	9.34	9.18	9.32	9.33	9.26	0.16
4)	9.33	9.32	9.32	9.34	9.26	9.33	9.33	9.31	9.24	9.25	9.303	0.1
5)	9.32	9.35	9.34	9.24	9.34	9.37	9.23	9.23	9.33	9.32	9.307	0.12

NORMAL DISTRIBUTION CURVE

Scale :-
X axis - 1 unit = 0·02
Y axis - 1 unit = 4

Normal Distribution of Sample Size :-

Theory :-

The Normal Distribution Curve is indicated and drawn using, histograms. It is plotted with the help of frequency of the sample mean in the given interval of value. The mid point of, histograms combined together and this results in a curve which follows a normal distribution curve.

Procedure :-

1) Select the quantity to be measured
2) Take suitable amount of indicated piece to take measurement
3) Grouping them in sample having a suitable amount of sample size.
4) Find out the mean of samples.
5) By collecting different sample mean, choose the interval
6) Find out the frequency of the individual interval in between the value of sample mean exist
7.) Draw the histogram
8.) Joint the mid point of the histogram.
9.) Analyze the curve.

Name.................................... Faculty No.................................... Class.................... Batch............

S.No.	Sample Elements										Average ($\bar{X}$)	Range (R)
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}		
6)	9.33	9.33	9.24	9.23	9.32	9.33	9.33	9.34	9.27	9.35	9.307	0.12
7)	9.34	9.32	9.32	9.32	9.33	9.33	9.26	9.24	9.32	9.32	9.31	0.1
8)	9.32	9.34	9.32	9.32	9.24	9.33	9.32	9.32	9.24	9.32	9.307	0.1
9)	9.19	9.17	9.33	9.33	9.26	9.32	9.33	9.33	9.33	9.32	9.29	0.16
10)	9.26	9.35	9.33	9.31	9.34	9.28	9.34	9.33	9.23	9.34	9.311	0.12
											$\bar{\bar{X}} = 9.301$	$\bar{R} = 0.121$

→ $\bar{X}$ Chart, $n = 10$, $\bar{\bar{X}} = 9.301$

$A_2 = 0.31$

$$U.C.L. = \bar{\bar{X}} + A_2\bar{R} = 9.301 + 0.31 \times 0.121$$

$$= 9.338$$

$$L.C.L. = \bar{\bar{X}} - A_2\bar{R} = 9.301 - 0.31 \times 0.121$$

$$= 9.26$$

→ R Chart, $n = 10$, $D_4 = 1.78$, $\bar{R} = 0.121$

$$U.C.L. = D_4 \times \bar{R} = 1.78 \times 0.121$$

$$= 0.21$$

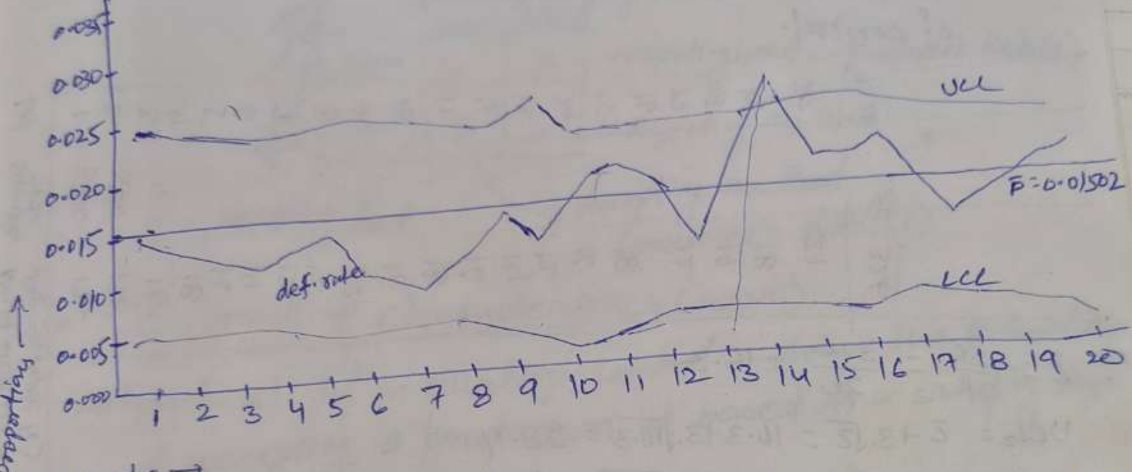
$$L.C.L. = D_3 \times \bar{R} = 0.22 \times 0.121$$

$$= 0.0266$$

taken.-

...lculate UCL + LCL. Since sample sizes are unequal, the control limits vary from sample interval to sample interval.

$$UCL_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}} \qquad LCL_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$



samples →

→ The proportion of defectives on day 13 is higher than the UCL. Therefore the process is out of control. Statisticians to identify the root cause for the cause + take appropriate corrective action to bring the process under control.

C-chart – Poisson

- control chart for defects, it can be single defect or multiple defects.

$$\bar{c} = \frac{\text{Total no. of defects}}{\text{No. of samples}} = \frac{\Sigma c}{k}$$

- used to track the no. of defects in a product of constant size.

$$UCL_c = \bar{c} + 3\sqrt{\bar{c}}$$
$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

- c = no. of defects
- k = no. of samples.

Binomial

P-chart

- Control chart for proportions
- Used to analyze the proportions of non-conforming or defect items in a process
- Used to detect unexpected changes in the process.
- Used to monitor process stability over time.
- Compare process performance before + after significant process improvements.

$\bar{n} = \frac{\Sigma n}{k}$

$\bar{p} = \frac{\Sigma np}{\Sigma n}$

$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$

$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$

np = no. of defectives in the sample

k = no. of Lots

n = sample size

Ex- ABC mfg. produces thousands of tubes everyday. A quality inspector randomly draws samples for 20 days and reports the defective tubes for each sample size. Based on the given data, prepare the control chart for the fraction of defective + determine the process in ~~sta~~ statistical control.

Lot (k)	Sample (n)	np	def. Rate np/n	UCLp	LCLp
1	1250	18	0.01440	0.02534	0.00470
2	1300	15	0.01154	0.02514	0.00490
3	1350	13	0.00963	0.02495	0.00509
4	1200	16	0.01333	0.02555	0.00448
5	1050	8	0.00762	0.02628	0.00376
6	1050	6	0.00571	0.02628	0.00376
7	1200	18	0.01500	0.02555	0.00448
8	1100	14	0.01273	0.02602	0.00402
9	1000	22	0.02200	0.02656	0.00348
10	600	12	0.02000	0.02941	0.00012
11	1350	13	0.00963	0.02495	0.00509
12	1250	19	0.01520	0.02534	0.00470
13	1200	33	0.02750	0.02555	0.00448
14	1100	20	0.01818	0.02602	0.00402
15	1050	20	0.01905	0.02628	0.00376
16	950	20	0.02105	0.02686	0.00318
17	1560	22	0.01410	0.02426	0.00578
18	1150	17	0.01478	0.02578	0.00426
19	1230	19	0.01545	0.02542	0.00461
20	1100	21	0.01909	0.02602	0.00402
$\Sigma k = 20$	$\Sigma n = 23040$	$\Sigma np = 346$			

$\bar{n} = \frac{23040}{20} = 1152$

$\bar{p} = \frac{\Sigma np}{\Sigma n} = \frac{346}{23040} = 0.01502$

$1 - \bar{p} = 0.98498$

... are unequal, the

Observations :-

S.No.	Interval	Tally	Frequency
1	9.16 - 9.18	II	2
2	9.18 - 9.20	IIII	4
3	9.20 - 9.22	I	1
4	9.22 - 9.24	~~IIII~~ ~~IIII~~ III	13
5	9.24 - 9.26	~~IIII~~ II	7
6	9.26 - 9.28	III	3
7	9.30 - 9.32	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ I	26
8	9.32 - 9.34	~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~ ~~IIII~~	40
9	9.34 - 9.36	III	3
10	9.36 - 9.38	I	1
11.	9.28 - 9.30		0
			100

Result :-

As per $\bar{x}$ and $\bar{R}$ chart in sample size - 5, all points are under the U.C.L. and L.C.L. This is because all the readings are quite correct.

For sample size - 10, all points are under UCL and L.C.L. Though the normal distribution curve drawn here is not uniform.

① Single Sampling Plan

- Random sample size (n) taken from total lot size (N)
- Acceptance number (C)

→ If $n \le C$ → Accept, $n > C$ → Reject Lot.

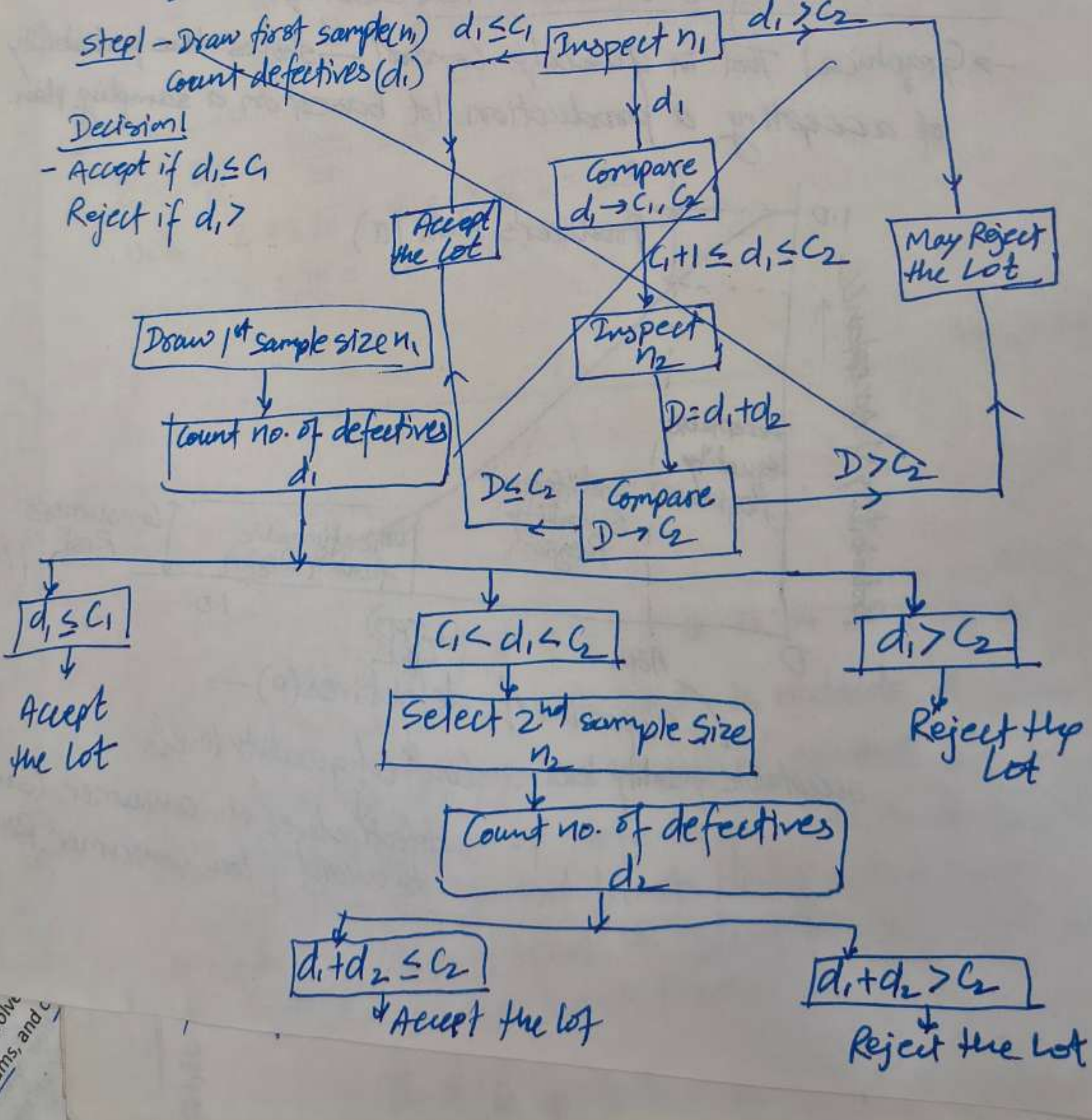
② Double Sampling Plan

- Acceptance or Rejection based on two samples.
- If 1st sample is Inconclusive → second sample is taken to make a final decision

→ n_1 → first sample size | C_1 → first acceptance no.
n_2 → second — — | C_2 → second "

step 1 – Draw first sample (n_1) count defectives (d_1)

Decision!
- Accept if $d_1 \le C_1$
Reject if $d_1 >$



Acceptance Sampling

- Used to decide whether to accept or reject an entire lot of materials based on testing random samples

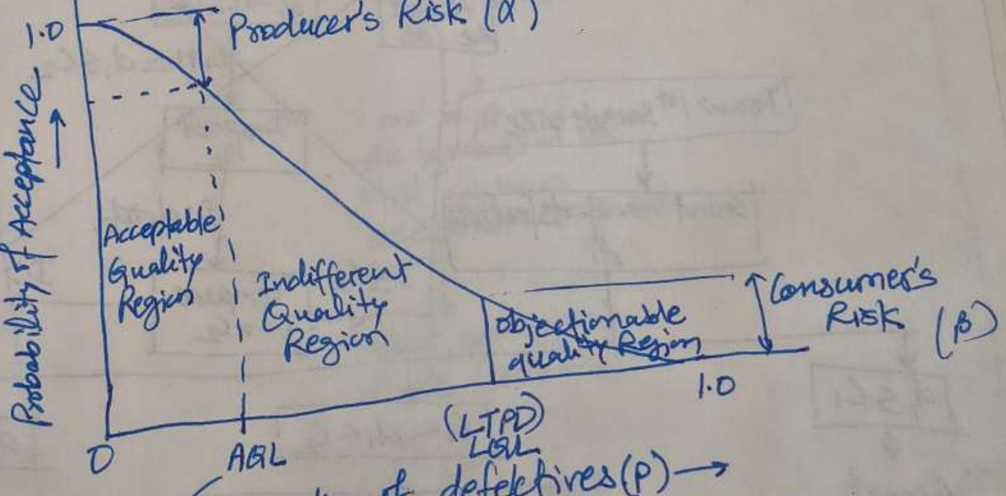
Actual quality of Incoming Product	Acceptance Sampling Results: Accept	Acceptance Sampling Results: Reject	
Good	✓	X	→ Rejecting Good – Producer's Risk (α)
Bad	X	✓	– Rejecting the Bad

Accepting Good

→ Accepting the Bad
Consumer's Risk. (β)

OC (operating characteristics Curve)

→ Graphical Tool in quality control — shows the probability of accepting a production lot based on a sampling plan.



Proportion of defectives (p) →

acceptable qulality level

Limited quality level

defective level the consumer wants to avoid (low consumer Risk)

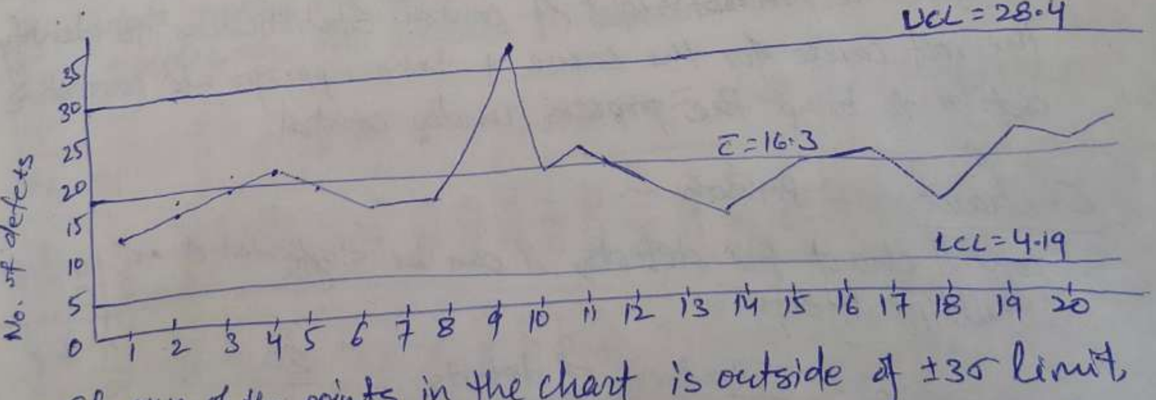
Ex- Mobile charger supplier drawn randomly constant sample size of 500 chargers everyday for quality control test. Defects in each charger are recorded during testing. Based on the given data, draw the appropriate control chart and comment on the state of control.

Lot	Sample size	No. of defects in the sample (c)
1	500	12
2	500	14
3	-	16
4	-	18
5	-	16
6	-	14
7	-	12
8	-	12
9	-	32
10	-	16
11	-	18
12	-	16
13	-	14
14	-	12
15	-	16
16	-	18
17	-	12
18	-	19
19	-	18
20	-	21
$k=20$		$\Sigma c = 326$

$\bar{c} = \frac{\Sigma c}{k} = \frac{326}{20} = 16.3$

$UCL_c = \bar{c} + 3\sqrt{\bar{c}} = 16.3 + 3\sqrt{16.3} = 28.4$

$LCL_c = \bar{c} - 3\sqrt{\bar{c}} = 16.3 - 3\sqrt{16.3} = 4.19$



- If any of the points in the chart is outside of $\pm 3\sigma$ limit, then consider the process is out of control. In the above example, the avg no. of defects per lot is 16.3, Sample 9 is outside of the control limit. Hence the process is out of control. Thus team needs to identify the root cause for the special cause variation.

Thickness of chip

$t_c \rightarrow$ chip thickness

$t \rightarrow$ uncut or undeformed chip thickness

$$r = \frac{t}{t_c} \qquad r < 1 \qquad t < t_c$$

t_c is increased due to shear failure

$t_c > t$ becoz of shear failure.

- There is no difference b/w depth of cut & uncut chip thickness in orthogonal machining only. ⇒ ($t = d$)
- For turning $t = d/\sin\lambda$

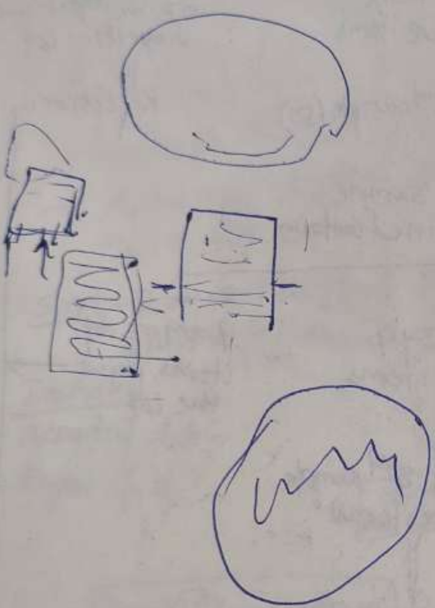
Rake Surface :- The surface along which chip moves upward is called rake surface of tool.

Flank or flank surface :

The other surface which is relieved to avoid rubbing with the machined surface is called flank or flank surface.

Rake angle -

Angle of inclination of rake surface from reference plane i.e normal to horizontal machined surface.



③ Multiple Sampling Plan

→ Typically → 3 to 7 or 8 → Make lot acceptance decisions → Ultimately reducing total no. of items inspected.

Inspection of first sample of (50 items)

1 or less no. of defective items accept the inspection Lot
Acceptance

B/w one & six defective items
Inspr of 2^{nd} sample (50)
if 1^{st} + 2^{nd} sample combined contains

Six or more defective items, reject the inspection Lot
Rejection

3 or less def. items, accept the lot

b/w 3 & 9 def. items
Inspect 3^{rd} sample (50 items)
and so on - if the Inspection lot still not been accepted or rejected.
Inspect last (eight) sample (50 items)
If all samples combined contains

9 or more def. items, Reject the lot

one or less no. of def. items, Accept the lot

12 or more def. items. Reject the lot

Negative Rake angle are recommanded-

- Maching high strength alloy.
- High speed cutting
- With rigid set-up
- Cutting tool material – Ceramic carbide (comp; (comp) (brittle)

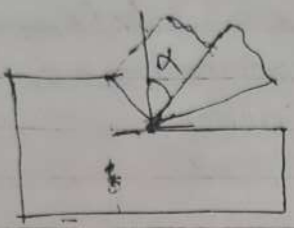
Zero Rake.

- To simplify design and manufacturing of the form of tool.
- Increase tool strength Tool strength ↑
- Avoids digging of tool into workpiece.
- **Brass** is turned with zero rake angle
- CI used zero rake angle.

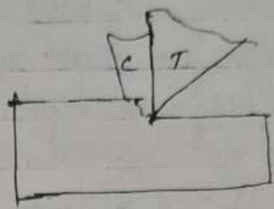
$\boxed{\text{Brass} + CI \rightarrow \alpha = 0}$

Discussion on Rake angle.

- Rake angle can be +ve, -ve or zero



$\alpha = 0$



Positive Rake. → [sharpness more]

- Reduce cutting force
- Reduce cutting power
- ~~Sh~~ Positive Rake angle are recommanded for-
- Maching low strength material.
- Low power maching
- Long shaft of small diameter
- set up lacks strength and rigidity.
- Low cutting speed.

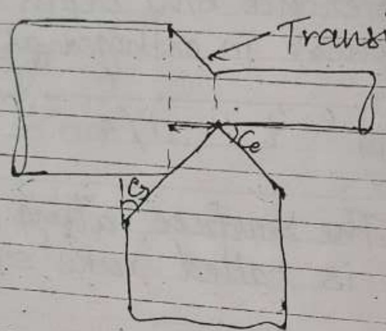
VIMP Cutting tool Material → High speed steel
18% Tungeston, 4% Cr, 1% V

Negative Rake.

- Increase edge strength (Mechanical & thermal)
- Increase life of the tool. Tool life ↑
- Icrease the cutting force
- Requires high cutting speed.
- Requires Ample (very high) power.
- Heavy impact loads.

Clearance Angle or relief angle (γ):
Angle of inclination of clearance or flank surface from the finished (machined surface)

<u>Turning</u>:-
Cutting Cond^n for turning.



· Speed $V = \pi D N$ mm/min
$= \frac{\pi D N}{60}$ mm/s.
$= \frac{\pi D N}{1000} = $ m/min.

· Feed is used for position the tool.
$f = $ mm/rev.

d = depth of cut (DOC) $= \frac{D_1 - D_2}{2}$

$$\boxed{D_2 + 2d = D_1}$$

System of Description of Tool Geometry.

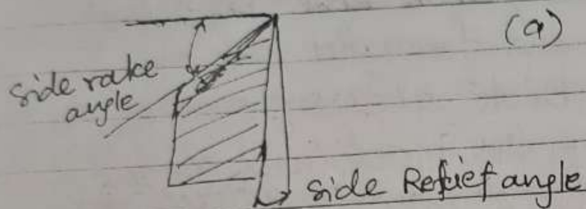
→ 1. M/c reference system - ASA or ANSI

→ 2. Tool reference system - ORS (ISO old) and NRS (ISO New)

→ 3. Work reference system - WRS.

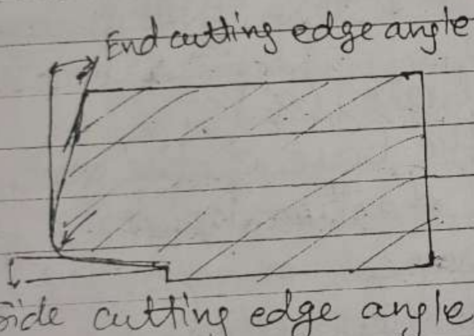
ASA (Single point cutting tool) , for description

(a)



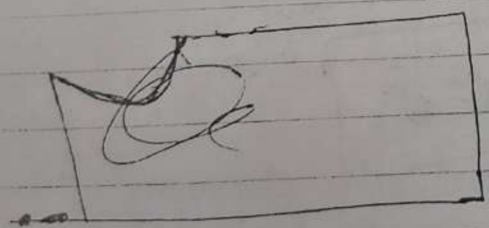
(Parallel to the width & perpendicular the base)

(b)

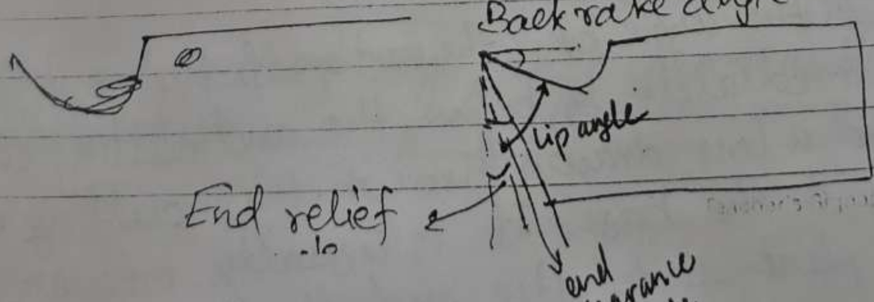


Parallel to the base

(C)



Perpendicular the base
& Parallel to the length



Orthogonal Cutting

1. Cutting edge of the tool is perpendicular to the dir^n of cutting velocity.
2. The cutting edge is wider than the workpiece width and extends beyond the workpiece on either side. Also the width of the workpiece is much greater than the depth of cut
3. The chip generated flows on the rake face of the tool with chip velocity ⊥ to the cutting edge.
4. The cutting forces act along two dir^{n}s only. (2D).

Geometry of single point ~~cutting~~ Turning tool: -

classification - (According to the no. of major cutting edges (points) involved.).

→ Single point -

Turning, shaping, planning, slotting tools, Parting tools etc. (vertical axis shaper)

→ Double point -

Drilling tool

→ Multipoint -

Milling, broaching, hobbing tools saw, grinding wheel etc.

Clearance angle :— 5–15°

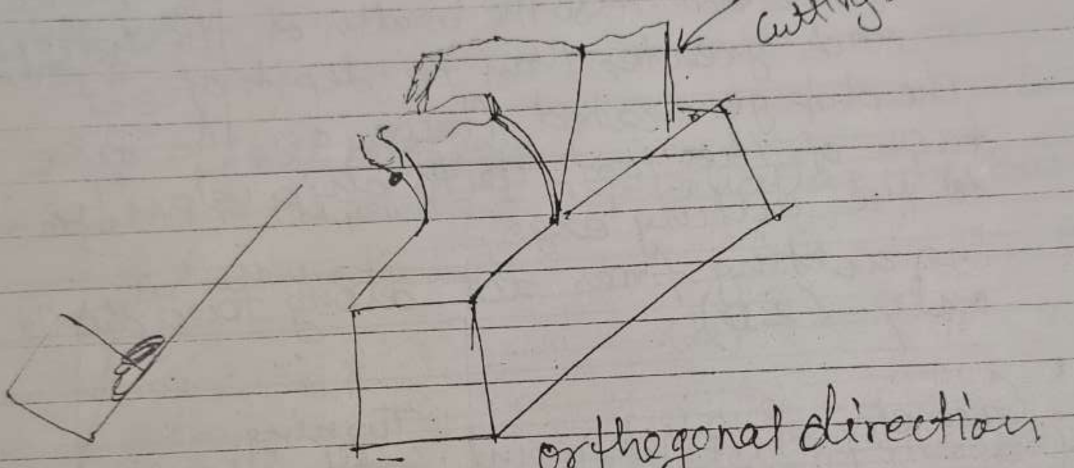
→ Provided to avoid rubbing of the tool (flank) with the machined surface.

→ Reduce tool wear

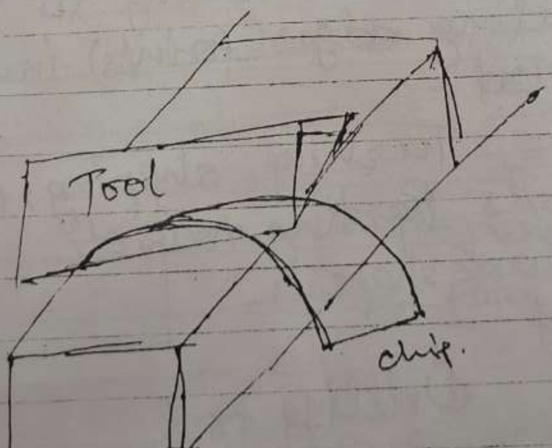
→ Increase tool life. Tool life ↑, wear ↓

→ Must be positive (3° – 15°)

Types of Machining :

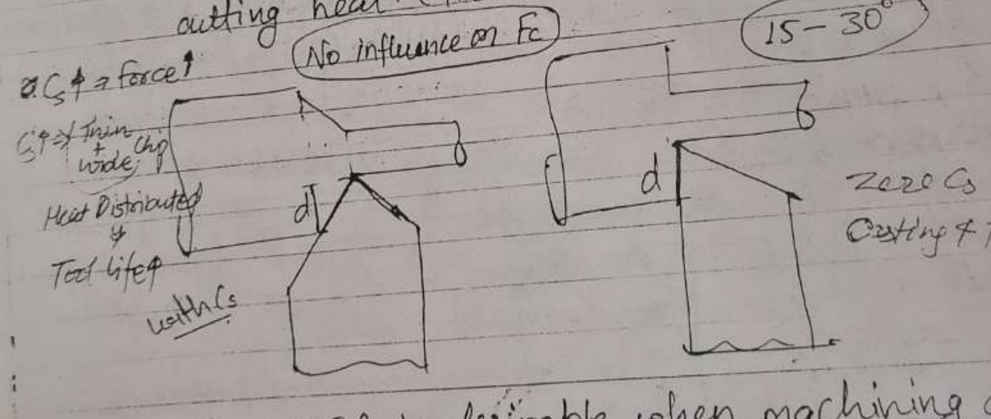


orthogonal direction



Oblique Cutting

VImp Side cutting edge angle, SCEA (C_s)
- It is the angle which prevents interference as the tool enters the work material.
(Normally 15-30°).
- Larger this angle, the greater the component of force tending to separate the work and the tool. (May induce chatter).
- At its increased value it will have more of its length in action for a given depth of cut
- At its increased value it produce thinner and wider chip that will distribute the cutting heat. (increase tool life)



- Zero SCEA is desirable when machining casting & forging with hard and scaly skins, because of the least amount of ~~the~~ tool edge should be exposed to the destructive action of the skin.
- SCEA has no influence on cutting force or F_z (cutting) cutting power consuption.

→ End relief angle prevents friction on the flank of the tool.

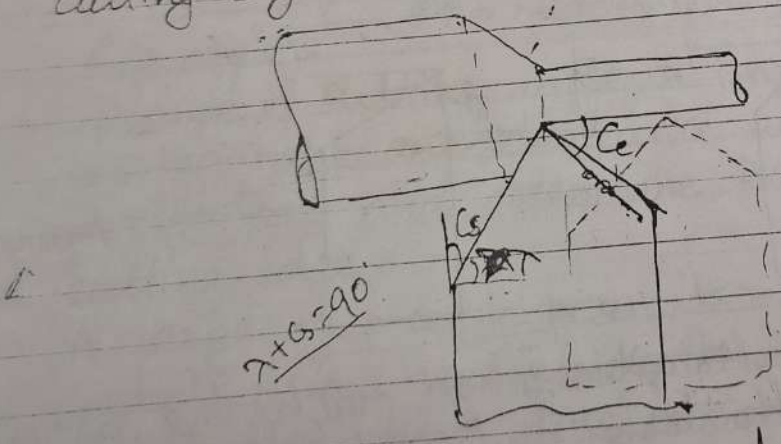
Side relief angle (γ_s)

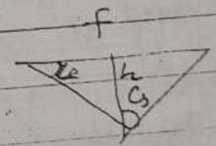
→ It is the angle b/w the portion of the side flank immediately below the side cutting edge, and a line drawn through this cutting edge ⊥ to the base.

→ It is measured in a plane ⊥ to the side flank.

End cutting edge angle, ECEA (C_e) —

End cutting edge angle is the amount that the
→ The end cutting edge angle
end-cutting edge slopes away from the nose of the tool, so that it will clear the finished surface on the workpiece, when cutting with the side cutting edge.





$C_e \uparrow \Rightarrow$ Weak Tool.

8–15°

→ It prevents the trailing end of the cutting edge of the tool from rubbing against the work-piece.

→ A larger end cutting edge angle weakens the tool.

→ It's usually kept b/w 8° to 15°

Back rake angle (α_b) —

→ It is the angle b/w the face of the tool and the base of the shank or holder & is usually measured in a plane ⊥ to the base & ∥ to the length of the tool.

→ It affects the ability of the tool to shear the work material and form the chip

→ In turning positive back rake angle takes the chips away from the machined surface whereas negative back rake angle direct the chip on to the machined surface.

Side rake angle (axial rake) (α_s) —

→ It is the angle b/w the base of the tool and the base of the shank or holder, and is usually measured in a plane ⊥ to the base & ∥ to the width.

→ Increase in the side rake angle reduces the chip thickness, turning.

$$\boxed{\alpha_s \uparrow \Rightarrow t_c \downarrow \qquad \uparrow r = \frac{t}{t_c \downarrow}}$$

End relief angle (γ_e)

→ It is the angle b/w the portion of the end flank immediately below the end cutting edge and a line drawn through this cutting edge ⊥ to the base. It is usually measured in a plane ⊥ to the end flank.

For orthogonal cutting, $i = 0$
For oblique cutting, $i \neq 0$

Inter-conversion b/w ASA & ORS. (IES (Generally).)

$\boxed{\tan\alpha = \tan\alpha_s \sin\lambda + \tan\alpha_b \cos\lambda}$ Imp.

$\tan\alpha_b = \cos\lambda \tan\alpha + \sin\lambda \tan i$
$\tan\alpha_s = \sin\lambda \tan\alpha - \cos\lambda \tan i$
$\tan i = -\tan\alpha_s \cos\lambda + \tan\alpha_b \sin\lambda$

Critical correlations:-

When $\lambda = 90$ $\qquad \alpha_s = \alpha$ ⇒ Imp.
When $i = 0$ $\qquad \alpha_n = \alpha$
when $i = 0$ & $\lambda = 90$ $\qquad \alpha_s = \alpha_n = \alpha$ ⇒ $C_s = C$
(Pure orthogonal cutting).

λ is principal cutting edge angle (or) Approach Angle
i is inclination angle
α_n normal rake angle (NRS)
α_s is side rake angle (ASA)
α is orthogonal rake angle (ORS)

Shear angle (ϕ):- cutting parameter

$$r = \frac{t}{t_c} = \frac{l_c}{l} = \frac{V_c}{V} = \frac{\sin\phi}{\cos(\phi - \alpha)} = \frac{1}{h}$$

and $\boxed{\tan\phi = \dfrac{r\cos\alpha}{1 - r\sin\alpha}}$ Imp.

→ It provide better surface finish.

$$\boxed{h\downarrow = \frac{f^2}{8R\uparrow}} = \frac{f}{\tan C_s + \cot C_e}$$

→ But tool large a nose radius will induce chatter.
→ If nose radius increased cutting force and cutting power increased.
• It increases the life of the tool.

Tool designation or Tool signature (ANSI)
or ASA

• To remember easily follow the rule
4 Rake, relief, cutting edge
→3 finish with nose radius (inch)
→2 Side will come last.

$$\boxed{\alpha_b - \alpha_s - \gamma_e - \gamma_s - C_e - C_s - R}$$

Orthogonal Rake System (ORS), ~~for~~ for calculation

$$\boxed{i - \alpha - \gamma - \gamma_1 - C_e - \lambda - R}$$

λ ↓ Greater than 60°

- Inclination angle (i)
- Orthogonal rake angle (α)
- Side relief angle (γ)
- End relief angle (γ_1)
- End cutting edge angle (C_e)
- Principal cutting edge angle or Approach angle ($\lambda = 90 - C_s$)
- Nose radius (R) (mm)

(Angle in drill generally 118° $\angle$.

F_y or radial force will increase with SCEA.

$$F_y\uparrow = f_t \cos\lambda = f_t \sin C_s \uparrow$$

F_x or Feed force will decrease with SCEA

$$F_x\downarrow = f_t \sin\lambda = f_t \cos C_s \uparrow \quad (\lambda + C_s = 90^\circ)$$

$\lambda \rightarrow$ principle cutting edge angle.

SCEA and Lead angle

Lead angle equal to ~~SE~~ SCEA

$\psi = C_s$

depth of cut

$\psi\uparrow \Rightarrow$ hard metals, $d\uparrow$, Tool life$\uparrow$.

Lip angle - or wedge angle.

→ Lip angle or cutting angle or knife angle depends on the rake and clearance angle provided on the tool and determine the strength of the cutting edge.

→ A larger lip angle permits machining of harder metals, allow heavier depth of cut, increase tool life, better head dissipation.

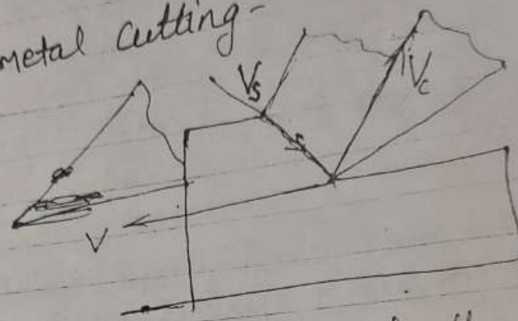
Nose Radius –

→ It ~~is~~ is curvature of the tool tip.

→ It strengthen the tool nose by reducing stress concentration. $K_t\downarrow$

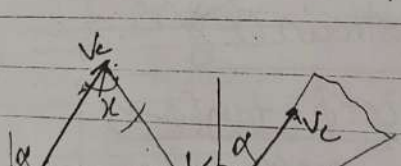
$\varepsilon_{min} = 2$ when $\phi = 45°$

Velocities in metal cutting-



- The velocity of the tool relative to the work piece (V) is called the cutting speed.
- The velocity of the chip relative to the work piece (V_s) is called the shear velocity
- The velocity of the chip relative to the tool (V_c) is called the chip velocity.

Derive the expression for velocities in metal cutting.



$$x + \phi + 90 - \alpha = 180$$

or $x = 90 - \phi + \alpha$

$$x = 90 - (\phi - \alpha)$$

Sine Rule

$$\frac{V}{\sin(90-(\phi-\alpha))} = \frac{V_c}{\sin\phi} = \frac{V_s}{\sin(90-\alpha)}$$

$$\boxed{\frac{V}{\cos(\phi-\alpha)} = \frac{V_c}{\sin\phi} = \frac{V_s}{\cos\alpha}}$$

$\sin\phi = r\cos(\phi-\alpha)$

$\sin\phi = r\cos\phi\cos\alpha + r\sin\phi\sin\alpha$

$\sin\phi - r\sin\phi\sin\alpha = r\cos\phi\cos\alpha$

or $\sin\phi(1-r\sin\alpha) = r\cos\phi\cos\alpha$

or $\dfrac{\sin\phi}{\cos\phi} = \dfrac{r\cos\alpha}{1-r\sin\alpha}$

or $\boxed{\tan\phi = \dfrac{r\cos\alpha}{1-r\sin\alpha}}$

A single-point cutting tool with 12° rake angle is used to m/c a steel work piece. Depth of cut i.e uncut thickness is 0.81 mm. The ch.

$t_c = 1.8$ mm , $\phi = ?$

$t = 0.81$ $\quad r = \dfrac{t}{t_c} = \dfrac{0.81}{1.8} = 0.45 = \dfrac{\sin\phi}{\cos(\phi-\alpha)}$

$t_c = 1.8$

$\tan\phi = \dfrac{0.45\cos 12}{1-0.45\sin 12}$ $\quad \phi = 26°$

Cutting shear strain (ε)

$$\boxed{\varepsilon = \cot\phi + \tan(\phi-\alpha) = \dfrac{\cos\alpha}{\sin\phi\cos(\phi-\alpha)}}$$

If $\alpha = 0$, then $\varepsilon = \cot\phi + \tan\phi \geq 2$ $\quad | \; x + \frac{1}{x}$ form.

$AM \geq GM$

$\dfrac{x+\frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}}$

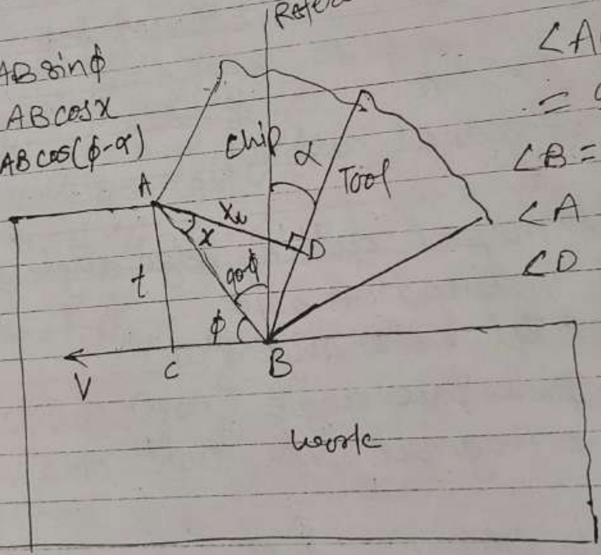
or $x + \frac{1}{x} \geq 2$

where
r = Chip thickness ratio or cutting ratio ; $r<1$
$h = 1/r$ = Inverse of chip ratio or chip reduction factor or chip compression ratio
$h > 1$.

- If $\underline{\alpha = 0}$; $\boxed{\tan\phi = r}$

$\underline{\text{Proove}}$: $\tan\phi = \dfrac{r\cos\alpha}{1 - r\sin\alpha}$ [IAS Main 2009

$t = AB\sin\phi$
$t_c = AB\cos x$
$t_c = AB\cos(\phi - \alpha)$



$\angle ABD$
$= 90 - \phi + \alpha$
$\angle B = 90 - \phi + \alpha$
$\angle A = x$
$\angle D = 90.$

$$180 = 90 - \phi + \alpha + x + 90$$

$$\underline{x = \phi - \alpha}$$

$$r = \frac{t}{t_c} = \frac{AB\sin\phi}{AB\cos(\phi - \alpha)}$$

$$\boxed{r = \frac{\sin\phi}{\cos(\phi - \alpha)}}$$

feed - large   [B/w top + bottom shear stress is more]
tool rake - negative
cutting fluid - absent or inadequate
Using multipoint cutting tool like milling, broaching.

→ Conditions for forming Continuous chip without BUE
- Work material - ductile
- Cutting velocity - high
- Feed rate - Low
- Rake angle - positive and high
- Cutting fluid — both cooling and lubrication.

→ Conditions for forming Continuous Chip with BUE
- Work material — ductile
- Cutting velocity — Medium
- Feed — Medium
- Cutting fluid - absent or inadequate.

→ Built-Up-Edge (BUE) formation -
- In machining ductile material with long chip tool contact length, lot of stress and temperature develops in the secondary deformation zone at the chip tool interface.
- Under such high stress & temp in b/w two clean surfaces of metals, strong bonding may locally take place due to addition similar to welding.
- Such bonding will be encouraged and accelerated if the chip tool materials have mutual affinity or solubility.

$$\tan\phi = \frac{r\cos\alpha}{1-r\sin\alpha}$$

$$= \frac{0.4\cos 10}{1-0.4\sin 10} = 0.42$$

$$\phi = 22.9^\circ \simeq 23^\circ$$

$$V_s = \frac{V\cos\alpha}{\cos(\phi-\alpha)}$$

$$= \frac{2.5\cos 10}{\cos 13^\circ} = 2.52$$

$$\dot{\epsilon} = \frac{V_s}{t_s} = \frac{2.52}{25\times10^{-6}} = 1.0\times10^{5}\ \frac{1}{sec}$$

Cause of chip formation-

- Shear yielding – in ductile material
- Brittle fracture – in brittle material

Types of chip.

- continuous chip
- Discontinuous chip
- Continuous chip with BUE
- Serrated chip

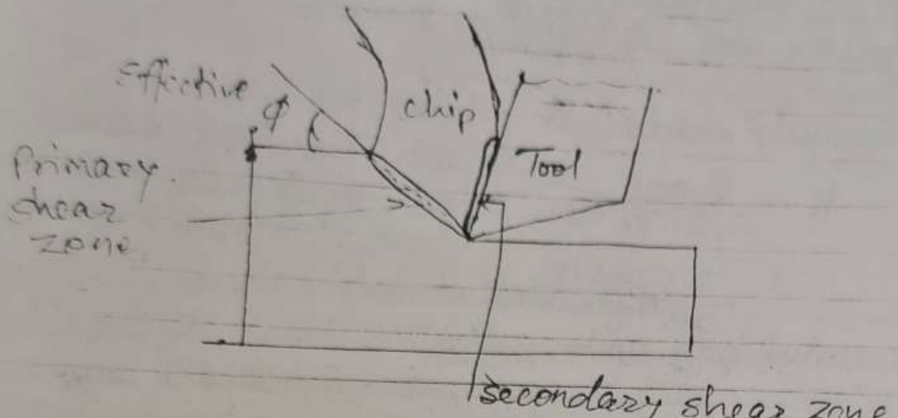
→ Conditions for forming Discontinuous chip.
of irregular size and shape
work material brittle (such as gray cast iron)

of regular size and shape
work material - ductile but hard and work hardnable

[illegible] carbon steel like mild steel

$$\frac{V_c}{V} = \frac{\sin\phi}{\cos(\phi-\alpha)} = r$$

$$\frac{V_s}{V} = \frac{\cos\alpha}{\cos(\phi-\alpha)}$$



Shear strain Rate.

(Note: It is not shear strain it is rate of shear strain i.e. flow)

$$\dot{\varepsilon} = \frac{d\varepsilon}{dt} = \frac{V_s}{\text{thickness of shear zone }(t_s)}$$

- Thickness of shear zone can be taken as 1/10th (10% of shear plane length and its max^m value is 25 micron
- Unit 1/sec.

Q. $r = 0.4$, $t = 0.6mm$, $\alpha = 10^\circ$, $V = 2.5\ m/s$, $t_s = 25\mu$.

$0.4 = \frac{V_c}{0.6} \Rightarrow \cancel{t_c = \frac{0.6}{0.4} = 1.5}$ $\dot{\varepsilon} = ?$

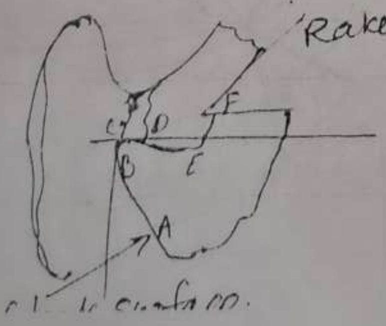
$\cancel{V_c = 0.24}$

[Titanium Low weight about $\frac{1}{2}$ of iron]
1.7

Decrease
Feed ↓
Depth of cut ↓
Use
- Cutting fluid
- Change cutting tool material (as cerments).

→ Serrated Chips. (semi continuous). → Ti K↓
- Serrated chips also called segmented or non-homogeneous chips are semi continuous chips with zones of low and high shear strain.
- Metals with low thermal conductivity & strain strength that decreases sharply with temp., such as titanium exhibit this behaviour, the chips have sawtooth like apperance.

When is forced chip breaking necessary and why?
When chips continuously form and come out very hot, sharp and at quite high speed.
Under the condition
- Soft ductile work material
- Flat rake surface with positive or near zero rake.
- For
- Safety and convenience of the operator.
- Easy collection and disposal of chips.



EF Chip breaker
[Integrated chip breaker

- The weld material starts forming as an embryo at the most favourable location and thus gradually grows.
- With the growth of the BUE, the force also gradually increase due to wedging action of the tool tip along with the BUE formed on it
- Whenever the force exceeds the bonding force of the BUE, the BUE is broken or sheared off and taken away by the following chip. The again starts forming and grow.
- Low cutting speed also contributes to the formation of BUE.
  (Milling, Broaching are prone to BUE formation due to low speed).

Effects of BUE formation:

Harmful effect

- Poor surface finish.
- It unfavourably changes the rake angle at the tool tip causing increase of cutting force i.e power consuption.
- Induce vibration. (dimentions & accuracy affected.

Good effect

- BUE protects the cutting edge of the tool i.e. increase tool life.

Reduction or Elimination of BUE

Increase

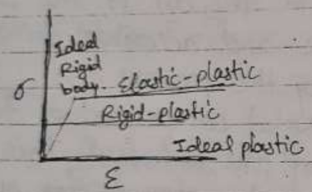
Cutting speed ↑

Rake angle ↑

If $\alpha = 0$ & $\beta = 45^\circ$
in merchant circle gives square
then $\boxed{F_c = F_t = F = N}$

$\alpha = 0, \beta = 45^\circ$ i.e. $\mu = 1$

Merchant Analysis - or Merchant Theory.
Assumption. different from merchant cycle
- The work material behaves like an ideal plastic.



- The theory involves minimum energy principal.
- τ_s and β are assumed to be constant, independent of ϕ.
- It is based on single shear plane theory.

From Merchant Analysis –

$$\boxed{\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \frac{\beta}{2}}$$

(deg.) (rad.) (Deg.) (Deg.)

$\phi_{\text{merchant theory}} \neq \phi_{\text{theoret.}}$

$F_{c\,\text{merchant}} < F_{c\,\text{theoretical}}$

F_c is less than actual force (F_c) as in theoretical

$$\phi = 45 + \frac{\alpha}{2} - \frac{\beta}{2}$$

$$2\phi = \frac{\pi}{2} + \frac{\alpha}{2} - \beta$$

$$\boxed{2\phi + \beta - \alpha = \frac{\pi}{2}}$$

Ans $\alpha = 10$ $\quad \tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$ $\quad r = \frac{t}{t_c} = \frac{0.2}{t_c}$ / ESE

$b = 2\,mm$

$\tau_s = 400\,MPa$

Applying merchant theory.

$$\phi = 45 + \frac{\alpha}{2} - \beta/2$$

$$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.5) = 26.56.$$

(i) shear angle $\phi = 45 + \frac{10}{2} - \frac{26.56}{2} = 36.7^\circ$

(ii) $F_s = \frac{\tau_s b t}{\sin\phi} = \frac{400\,N}{mm^2} \times \frac{2\,mm \times 0.2\,mm}{\sin 36.7}$

$= 267.5\,N$

$$F_s = R\cos(\phi + \beta - \alpha)$$

or $267.5 = R\cos(36.7 + 26.56 - 10)$

or $R = 447.5\,N$

Cutting force $F_c = R\cos(\phi - \alpha)$ $\quad R\cos(\beta - \alpha)$
$= 447.5\cos(36.7 - 10) = 429\,N$

Thrust force: $F_t = R\sin(\beta - \alpha)$
$= 447.5\sin(36.7 - 10) = 127\,N$

Modified Merchant Theory:

$$\tau_s = \tau_{s0} + K\sigma_s = F_s/A_s + K\frac{F_N}{A_s}$$

where, σ_s is normal stress on shear plane $\left[\sigma_s = \frac{F_N}{A_s}\right]$

and then $\boxed{2\phi + \beta - \alpha = \cot^{-1}(k)}$

$F = R\sin\beta$

$N = R\cos\beta$

$F_c = R\cos(\beta-\alpha)$

$F_t = R\sin(\beta-\alpha)$

$F_s = R\cos(\phi+\beta-\alpha) = \tau_s \frac{bt}{\sin\phi}$

$F_N = R\sin(\phi+\beta-\alpha)$

Ans: $\alpha = 10°$ $\quad r = 0.35$ $\quad t = 0.51\ mm$ $\quad b = 3\ mm$

$\tau_s = 285\ N/mm^2$ $\quad \mu = 0.65$

$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.65) = 33°$

$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha} = \frac{0.35\cos 10}{1 - 0.35\sin 10}$

or $\phi = 20.15°$

$F_s = R\cos(\phi+\beta-\alpha) = \frac{\tau_s bt}{\sin\phi}$

$R\cos(20.15+33-10) = 285\frac{N}{mm^2} \times \frac{3\ mm \times 0.51\ mm}{\sin(20.15)}$

$F_s = 1265.8\ N$

or $R = \frac{1265.8}{\cos(20.15+33-10)} = 1735.3\ N$

(i) Cutting force $F_c = R\cos(\beta-\alpha)$

$= 1735.3\cos(33-10)$

$= 1608\ N$

[F = μN by Coulomb's law for sliding friction]

(ii) Radial force $= 0$ [Only in turning operation]

(iii) Normal force $(N) = R\cos\beta$

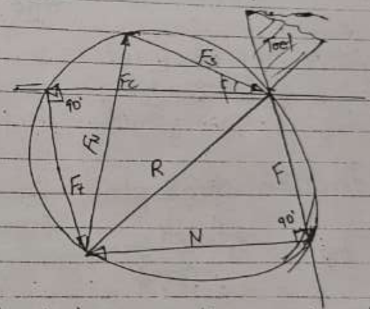
$= 1735.3\cos 33 = 1456\ N$

(iv) Shear force $(F_s) = 1265.8\ N$

Limitations of use of MCD

1. MCD is valid only for orthogonal cutting.
2. By the ratio F/N, the MCD gives apparent (not actual) co-efficient of friction.

If Rake angle $\alpha = 0$

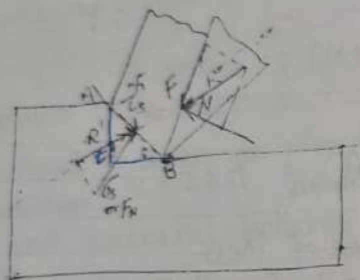


Merchant circle gives the rectangle

$$\boxed{\begin{array}{l} F_c = N \\ F_t = F \end{array}}$$

2. Force and Power in Metal Cutting

[For orthogonal metal cutting]



F = friction force
N = Normal force

$F = \mu N$

Length of shear plane (AB)

$t = AB \sin\phi$

or $AB = \dfrac{t}{\sin\phi}$

Area of shear plane $(A_s) = \dfrac{bt}{\sin\phi}$

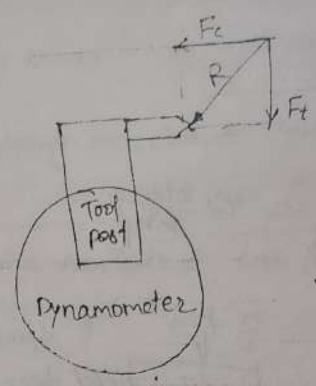
Shear ~~stress~~ Force on shear plane

$$F_s = \tau_s \times \frac{bt}{\sin\phi}$$

Normal force on shear plane

$$F_N = \sigma_s \times \frac{bt}{\sin\phi}$$

Chip is moving with constant velocity (V_c) therefore net resultant force on the chip must be zero. Therefore R and R' must be equal and opposite.

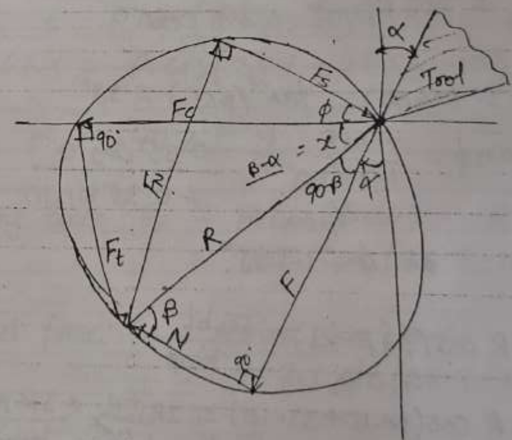


Cutting force or Horizontal force (F_c)

Thrust force or vertical force (F_t).

Merchant force circle diagram (MCD).

Resultant R as a diameter



$\tan\beta = \dfrac{F}{N} = \mu$

or $\beta = \tan^{-1}(\mu)$ = Friction angle

$x + 90 - \beta + \alpha = 90$

or $x = \beta - \alpha$

(iii) Cutting power

$P = F_c \times V$

$= F_c \times \frac{\pi D N}{60}$

$= 1200 \times \frac{\pi \times 0.160 \times 400}{60}$

$= 4021\ W$

$= 4.021\ kW$ Ans.

Ans. $D = 160\ mm$.

$d = 2.5\ mm$ $N = 315\ rpm$

$f = 0.16\ mm/rev$.

Tool geometry 0, 10, 8, 9, 15, 75, 0 mm (α = 10, λ = 75)

$F_c = 500N$

$F_t = \frac{F_x}{\sin\lambda} = \frac{200N}{\sin 75} = 207.05\ N$

$b = \frac{d}{\sin\lambda} = \frac{2.5}{\sin 75} = 2.58\ mm$

$t = f\sin\lambda = 0.16\sin 75 = 0.15\ mm$

$F = F_c \sin\alpha + F_t\cos\alpha$

$= 500\sin 10 + 207.05\cos 10 = 290.67\ N$

$N = F_c\cos\alpha - F_t\sin\alpha$

$= 500\cos 10 - 207.05\sin\alpha = 456.45\ N$

$F_s = F_c\cos\phi - F_t\sin\phi$

$\cancel{\tan\phi} = r = \frac{t}{t_c} = \frac{0.15}{0.48} = 0.312$

$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha} = \frac{0.312\cos 10}{1 - 0.312\sin 10} =$

$\phi = \tan^{-1}\ 18.43$

$F_s = 500\cos 18.43 - 207.05\sin 18.43$

$= \cancel{410.27\ N}\ 408.77\ N$

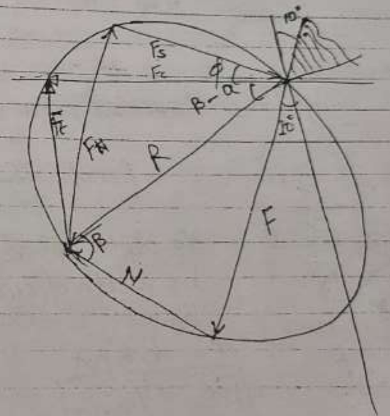
$F_N = F_c\sin\phi + F_t\cos\phi$

$= 500\sin 18.43 + 207.05\cos 18.43$

$= \cancel{351.37\ N}\ 354.64\ N$

$\tan\beta = \frac{F}{N} = \frac{290.67}{456.45}$

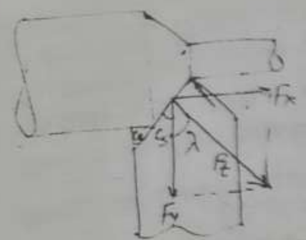
$\beta = 32.48°$



$F = R\sin\beta \Rightarrow R = \frac{F}{\sin\beta} = \frac{290.67}{\sin 32.48}$

$= 541.28\ N$

Proove:



$F_y = F_t \cos\lambda$ or $F_t = \dfrac{F_y}{\cos\lambda}$

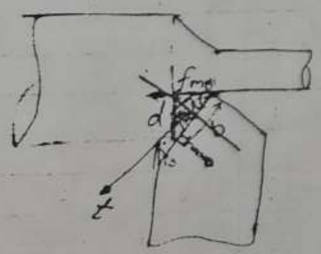
$F_x = F_t \sin\lambda$ or $F_t = \dfrac{F_x}{\sin\lambda}$

$F_t = \sqrt{F_x^2 + F_y^2}$

$F_t = \dfrac{F_x}{\sin\lambda} = \dfrac{F_y}{\cos\lambda} = \sqrt{F_x^2 + F_y^2}$

Geometry Conversion:
[Limited ~~only~~ to Orthogonal plane Only].

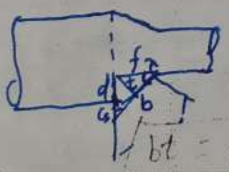
feed = f rev/mm



d = depth of cut
b = width of cut

$d = b \sin\lambda$

$\boxed{b = d/\sin\lambda}$



$\boxed{t = f \sin\lambda}$

$\boxed{bt = fd}$

Q. D = 160 mm Diameter
Tool geometry 0, 0, 10, 8, 15, 75, 0 (mm)
(α = 0, λ = 75)

$F_z = F_c = 1200\,N$

$F_x = F_t = 800\,N$

$F_t = \dfrac{F_x}{\sin\lambda} = \dfrac{800}{\sin 75} = 828\,N$

$b = \dfrac{d}{\sin\lambda} = \dfrac{4.0}{\sin 75} = 4.14\,mm$

$t = f\sin\lambda = 0.32 \sin\lambda = 0.3091\,mm.$

(i) $F = F_c \sin\alpha + F_t \cos\alpha$
$= 1200 \sin 0 + 828 \cos 0$
$= 828\,N$

$N = F_c \cos\alpha - F_t \sin\alpha$
$= 1200 \cos 0 - 828 \sin 0$
$= 1200\,N$

$\mu = \dfrac{F}{N} = \dfrac{828}{1200} = 0.69$

(ii) $F_s = F_c \cos\phi - F_t \sin\phi$ Yield shear strength?

$r = \dfrac{t}{t_c} = \dfrac{0.3091}{0.8} = 0.386$

$\alpha = 0$

$\tan\phi = r$, $\phi = \tan^{-1}(0.386) = 21.125^\circ$

$F_s = 1200 \cos 21.125^\circ - 828 \sin 21.125$
$= 821\,N$

$\tau_s = \dfrac{F_s}{A_s} = \dfrac{F_s}{bt/\sin\phi} = \dfrac{F_s \sin\phi}{bt}$

$= \dfrac{821 \sin 21.125}{4.14 \times 0.3091} = 231 \dfrac{N}{mm^2}$

Theory of Lee and Shaffer

- They applied the theory of plasticity for an ideal-rigid-plastic body.
- They also assumed that deformation occurs on a thin-shear plane.

And derive.

$$\boxed{\phi = \frac{\pi}{4} + \alpha - \beta}$$ $F_r \to F_{\text{frictional}}$

Other relations
- By stabber

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \beta$$

The force relations - (VIMP)

$F = F_c \sin\alpha + F_t \cos\alpha$
$N = F_c \cos\alpha - F_t \sin\alpha$
$F_N = F_c \sin\phi + F_t \cos\phi$
$F_s = F_c \cos\phi - F_t \sin\phi$

when F_c & F_t are known.

and $\mu = \frac{F}{N} = \tan\beta$

Turning Operation :- (Oblique Cutting) (~~Orthogonal Cutting~~).

Radial (axial force) feed force F_x; $V_f = fN$ mm/min

F_y Radial force (No work done) Thrust force

v cutting velocity

F_z: Cutting force (Tangential force)

$F_y = F_t \cos\lambda = F_t \sin\lambda$

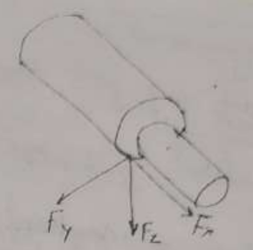
$F_x = F_t \sin\lambda = F_t \cos\lambda$

Feed

feed velocity $V_f = fN$ mm/min

$F_z \times V$: cutting power (about 99.4%)

$F_x \times V_f$: feed power (0.4 Negligible) very low



- F_c, or F_z or Tangential component force or primary cutting force acting in the dir^n of the cutting velocity, largest force and accounts for 99% of the power required by the process.
- F_t or F_x or axial component or feed force acting in the dir^n of the tool feed. This force is about 50% of F_c but accounts for only a small % of the power required because feed rates are usually small compare to cutting speeds.
- F_r or F_y or radial force or thrust force in turning acting $\perp$ to the machined surface. This force is about 50% of F_t i.e 25% of F_c and contributes very little to power requirements because velocity in the radial dir^n is negligible

Conversion Formula:

We have to convert turning (3D) to Orthogonal cutting (2D).

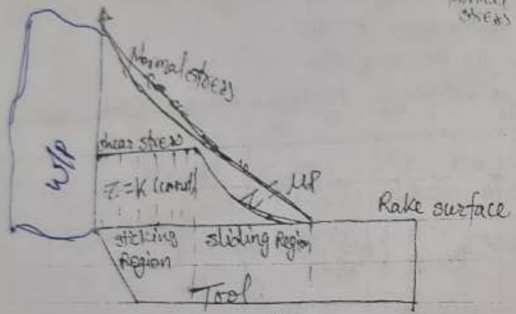
$$\boxed{F_c = F_z}$$

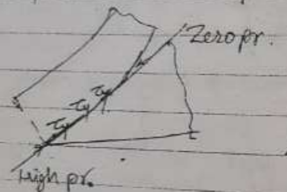
$$\boxed{F_t = F_{xy} = \frac{F_x}{\sin\lambda} = \frac{F_y}{\cos\lambda} = \sqrt{F_x^2 + F_y^2}}$$

Friction in Metal Cutting:

$\sigma = P$

Normal stress   Pressure





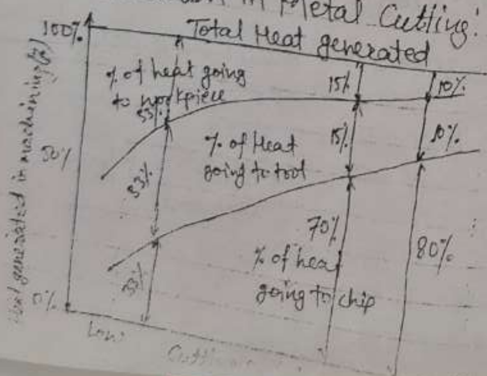
$F = \mu N$

$F = \mu N$

$F/A = \mu N/A$

$\boxed{\tau = \mu P}$ for sliding friction

Heat Distribution in Metal Cutting:



Determination of Cutting heat & temp. Experimentally

Heat

- Calorimetric method

Temperature

- Decolouring agent
- Tool-work thermocouple
- Moving thermocouple technique
- Embedded thermocouple technique
- Using compound tool
- Indirectly from hardness and structral transformation
- Photo-cell technique
- Infaray detection method.

Dynamometer:

- Dynamometers are used for measuring cutting forces.
- For ~~ot~~ orthogonal cutting use 2D dynamometer
- For oblique cutting use 3D dynamometer.

Types of Dynamometer

- Strain gauge type
- Piezoelectric type

Gate ~~2008~~(pi)

Carbon steel tools 250°C
Tongsten carbide tools 800°C
High speed steel tools 600°C.
Ceremic tools. 1200°C

Metal Removal Rate (MRR).

Metal Removal Rate (MRR) = $A_c \cdot V$ mm/min $\frac{mm^3}{min}$

$= btV$

$= fdV$

Where

A_c = Cross-section area of uncut chip (mm²)

V = Cutting speed = πDN, mm/min

Ques: $V = 50$ m/min $f = 0.8$ mm/rev $d = 1.5$ mm

$MRR = fdV = 0.8 \times 1.5 \times 50 \times 1000$ mm³/min

$= \cancel{60{,}000}$ mm³/min

$=$

Ques. $D = 200$ m $f = 0.25$ mm/rev $d = 4$ mm

$N = 160$ rpm

$$MRR = \frac{0.25 \times 4 \times \pi \times 200 \times 160}{60} \text{ mm}^3/\text{sec}$$

$= 1675.5$ mm³/sec

Power consumed During Cutting:

$$\boxed{P = F_c \times V} \text{ Watt}$$

(F_c in N, V in m/s)

where

F_c = Cutting force, (in N)

V = Cutting Speed $= \frac{\pi DN}{60}$, m/s

Total Power = Shearing Power + Friction Power

$$\boxed{F_c \times V = F_s \times V_s + F \times V_c}$$

60-75% 30-35%

% of Frictional Power $= \frac{F \times V_c}{F_c \times V} \times 100\%$

% of Shearing Power $= \frac{F_s \times V_s}{F_c \times V} \times 100\%$

Specific Energy Consuption (J/mm³):

Conventional Machine (Lathe, Drill, Milling) → 2-3 J/mm³

Grinding → 50 J/mm³

Electrochemical Machine (ECM) → 500 J/mm³

Specific Energy Consuption

$$\boxed{e = \frac{\text{Power (J/s)}}{\text{MRR (mm}^3/\text{s)}}}$$

$$\boxed{e = \frac{F_c \times V}{bt\,V}}$$ (V in m/s in the numerator, mm/s in the denominator)

$$\boxed{e = \frac{F_c}{1000\,bt}} = \frac{F_c}{1000\,fd}$$

Ques $e = 2.0$ J/mm³

$V_c = 120$ m/min $f = 0.2$ mm/rev $d = 2$ mm

$2 = \frac{F_c}{1000 \times 2 \times 0.2} \Rightarrow F_c = 800$ N

Ques. $D_o = 200$ m $D_i = 80$ m. $f = 0.1$ mm/rev $d = 1$ mm.

$V_c = 90$ m/min $F_c = 200$ N

$e = \frac{F_c}{1000\,fd} = \frac{200}{1000 \times 0.1 \times 1} = 2$ J/mm³

$\lambda = 60°$ $\quad i = 90 - \lambda$

7, 6, 6, 8, 30, 1 (mm) ASA

$t_o = 2.0\,mm$ $\quad b = 2.5\,mm$

$F_c = 1177\,N$ $\quad F_t = 560\,N$

$\tan\alpha = \tan\alpha_s \sin\lambda + \tan\alpha_b \cos\lambda$

$\tan\alpha = \tan\alpha_s \sin 60 + \tan 7 \cos 60$ —— (1)

$\tan\alpha_b = \tan\alpha \cos\lambda + \sin\lambda \tan i \nearrow 0$

$\tan 7 = \tan\alpha \cos 60$

$\alpha = 13.8°$

from (1)

$\tan 13.8 = \tan\alpha_s \sin 60 + \tan 7 \cos 60$

or $\alpha_s = 12°$

(i) side rake angle $\alpha_s = 12°$

(ii) $\mu = ?$ $\quad F = F_c \sin\alpha + F_t \cos\alpha$

$= 1177 \sin 13.8 + 560 \cos 13.8$

$= 824\,N$

$N = F_c \cos\alpha - F_t \sin\alpha$

$= 1177 \cos 13.8 - 56 \sin 13.8$

$= 1009\,N$

$\mu = \frac{F}{N} = \frac{824}{1009} = 0.816$

(iii) Dynamic shear strength of the work material

$\tau_s = \frac{F_s}{A_s}$

$F_s = F_c \cos\phi - F_t \sin\phi$

Applying merchant theory.

$\phi = 45 + \alpha/2 - \beta/2$

$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.816) = 39.2°$

$\phi = 45 + \frac{13.8}{2} - \frac{39.2}{2} = 32.3°$

$F_s = 1177 \cos 32.3° - 560 \sin 32.3$

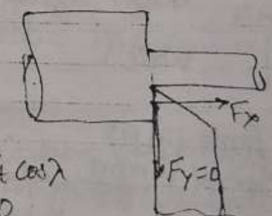
$= 696\,N$

$\tau_s = \frac{F_s}{A_s} = \frac{F_s}{bt/\sin\phi} = \frac{F_s \sin\phi}{bt}$

$= \frac{696\,N \sin 32.3}{2 \times 2.5\ mm.mm} = 74\ N/mm^2$

Orthogonal Turning —

$\boxed{\lambda = 90° \text{ i.e } C_s = 0}$



$F_y = F_t \cos\lambda$
$= 0$

$F_y = 0$

F_x & F_z exists.

$F_c = F_z$

$F_t = \frac{F_x}{\sin 90} = F_x$

$b = d$

$t = f$

$V_1 T_1^{0.25} = V_3 T_3^{0.25}$

$\cancel{350} \times 50 \times \left(\frac{2}{4}\right)^{0.25} = \cancel{350} \times 60 \times \left(\frac{x + \cancel{4}}{\cancel{4} \times 60}\right)^{0.25}$

or $\frac{50}{6} = 60 \times \left(\frac{x}{60}\right)^{0.25}$

or $\frac{5}{6} = \left(\frac{x}{60}\right)^{0.25}$

or $x = 60 \times \left(\frac{5}{6}\right)^4$

$= 28.935 \simeq 29$

2. IES 06 HSS Tool $T_1 = 1\,hr = 60\,min$

$V_1 = 30\,m/min$

$T_2' = 2.0\,min$

$V_2' = 60\,m/min$

$D = 300\,mm$. for $T_3' = 30\,min$

$V_1 T_1^n = \cancel{C}\ \cancel{V_2 T_2^n} = V_3 T_3^n$

$30 \times 60^n = C$

$\cancel{2} 60 \times 2^n = C$

$\cancel{k}\ 30 \times 60^n = \frac{60 \times 2^n}{2}$

$\frac{60}{2} = 2^{1/n}$

$30 = 2^{1/n}$

$\log 30 = \cancel{2} \frac{1}{n} \log 2$

$1.477 = \frac{1}{n} \times 0.3010$

$n = 0.2038$

$V_1 T_1^n = V_3 T_3^n$

$30 \times 60^{0.2038} = V_3 \cancel{\times} (30)^{0.2038}$

71

$V_3 = \cancel{3}\ 35\,m/min$

$V_3 = \pi D N \Rightarrow N = \frac{35}{\pi \times 0.3} = 37.13\,rpm$

Gate* Clearance angle changed from 10° to 7° (α).

[Flank] flank wear is proportion to $\cot\alpha$

rake angle zero tool life: ?

Tool life $\propto \frac{1}{\text{Tool wear}}$

$T \propto \frac{1}{\cot\alpha}$

$T \propto \tan\alpha$

$T = K \tan\alpha$

$T_1 = K \tan 10$

$T_2 = K \tan 7$

$\frac{T_2 - T_1}{T_1} \times 100\% = \frac{K \tan 7 - K \tan 10}{\tan 10} \times 100$

$= -30\%$

approx change in tool wear 30% decrease

IES 09 conventional $T_0 = \left(T_c + \frac{C_t}{C_m}\right)\left(\frac{1-n}{n}\right)$

$T_c = 3\,min$ $C_t = 3\,min \times 0.5 + 5 = Rs.\ 6.5/grind$

$C_m = Rs\ 0.50$ per min

$n = 0.2$

$T_0 = \left(3 + \frac{6.5}{0.50}\right)\left(\frac{1-0.2}{0.2}\right) = 64\,min$

$V_0 = \frac{C}{T_0^n} = \frac{60}{64^{0.2}} = 26\,m/min$

Given Cutting speed, m/min : 49.74, 49.23, 48.67, 45.76, 42.58
Tool life, min : 2.94, 3.90, 4.77, 9.87, 28.27

$c = ?$ $VT^n = c$ $\log V + n \log T = \log c$
$x + ny = K$

~~Bg~~ Linear Regration method is used

	V	T	$x=\log V$	$y=\log T$	xy	x^2
1	49.74	$T = 2.94$	$x_1 = 1.6967$	$y_1 = 0.4683$	~~1.8301~~	15.2..
2	49.23	3.90	$x_2 = 1.6922$	$y_2 = 0.5910$	1.000	
3	48.67	4.77	$x_3 = 1.6872$	$y_3 = 0.6785$	1.1447	
4	45.76	9.87	$x_4 = 1.6604$	$y_4 = 0.9943$	1.6509	
5	42.58	28.27	$x_5 = 1.6260$	$y_5 = 1.4502$	2.3580	

$\Sigma x = 8.36$ $\Sigma y = 4.18$ $\Sigma xy = 6.954$ $\Sigma x^2 = 14$

$y = ax + b$
$y_1 = ax_1 + b$
$y_2 = ax_2 + b$
$y_3 = ax_3 + b$
$y_4 = ax_4 + b$
$y_5 = ax_5 + b$
$\boxed{\Sigma y = a\Sigma x + b \cdot N}$

$y = ax + b$
$\Sigma y = a\Sigma x + b \times 5$
$4.18 = a \times 8.36 + b \times 5$ —— ①

$xy = ax^2 + bx$ | Multiplying by x both side.
$\Sigma xy = a\Sigma x^2 + b\Sigma x$
$6.954 = a \times 14 + b \times 8.36$ —— ②

By eqⁿ ① + eqⁿ ②
$a = -~~9.099~~ 14.28$
$b = 24.71$

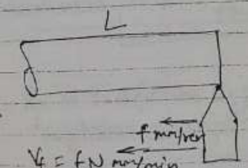
$y = ax + b$
$y = -14.28x + 24.71$
$VT^n = c$
$\log V + n \log T = \log c$
$x + ny = ~~K~~ \log c$
$y = -\frac{1}{n}x + \frac{1}{n}\log c$

$\frac{1}{n} = 14.28$ $\frac{1}{n}\log c = 24.71$
$n = \frac{1}{14.28} = 0.070$ $\log c = 1.730$
$c = 54$ Ans

Take 03

$V_1 = \pi D \times 50$ m/min
$V_2 = \pi D \times 80$ m/min
$V_3 = \pi D \times 60$ m/min

10 Tools produce 500 Comp.
1 Tool → 50 Com.

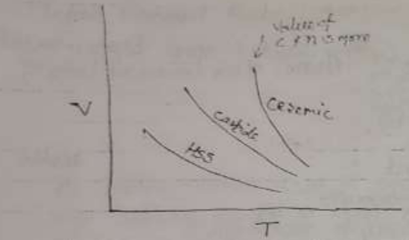


$V_f = fN$ mm/min
$t = \frac{L}{fN}$ min.

$T_1 = 50 \times t_1 = 50 \times \frac{L}{f \times 50} = \frac{L}{f}$ min
$T_2 = 12.2 \times t_2 = 12.2 \times \frac{L}{f \times 80}$
$T_3 = x \times \frac{L}{f \times 60}$

$V_1 T_1^n = V_2 T_2^n = V_3 T_3^n$
$\pi D \times 50 \times \left(\frac{L}{f}\right)^n = (\pi D \times 80) \times \left(12.2\frac{L}{f \times 80}\right)^n$
$50 = 80 \times \left(\frac{12.2}{80}\right)^n$
$\frac{5}{8} = \left(\frac{12.2}{80}\right)^n$ $\Rightarrow n = 0.25$

Tool life curve

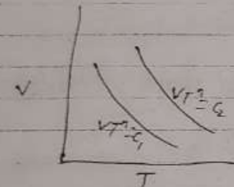


Taylor eqⁿ

$VT^n = C$

$yx^n = c$

or $y = \frac{c}{x^n}$

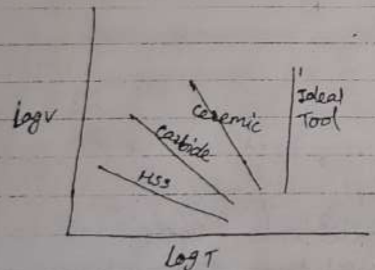


for same material (parallel)

$VT^n = C$

$VT^n = C_1 = 100$

$VT^n = C_2 = 200$

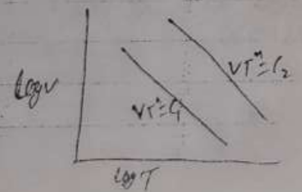


$VT^n = C$

$\log V + n \log T = \log C$

$y + nx = k$

$y = -nx + k$



for same material (||)

$C_2 > C_1$

Gate 2010

Q. for Tool A $n = 0.45$ $k = 90$

for Tool B $n = 0.3$ $k = 60$

$x\, T_A^{0.45} = 90 \Rightarrow T_A = \left(\frac{90}{x}\right)^{\frac{1}{0.45}}$

$x\, T_B^{0.3} = 60 \Rightarrow T_B = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$

$T_A > T_B$ $\quad \left(\frac{90}{x}\right)^{\frac{1}{0.45}} > \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$

or $\left(\frac{90}{x}\right)^{\frac{1}{0.45}} = \left(\frac{60}{x}\right)^{\frac{1}{0.3}}$

$x = 26.67$ m/min.

Gate 2013

Carbide Tool $VT^{1.6} = 3000$

HSS Tool $VT^{0.6} = 2000$

$\left(\frac{3000}{V}\right)^{\frac{1}{1.6}} > \left(\frac{2000}{V}\right)^{\frac{1}{0.6}}$

$\frac{1}{1.6} \log\left(\frac{3000}{V}\right) = \frac{1}{0.6} \log\left(\frac{2000}{V}\right)$

$0.625 \log\left(\frac{3000}{V}\right) = 1.66 \log\left(\frac{2000}{V}\right)$

$-3.7912 = \cancel{1.66\log\ 0.625\log V\ 1.66\log V}$

$= \cancel{1.035 \log V}$

$\log V = 3.6629$

$V = \cancel{38.9}\ 39.4$ m/min

38.9 m/min

5.1656 d − 1.6P

$h_0 = 4mm$ $D = 300mm$

$\mu = 0.1$

$h_{f\,min} = h_0 - \mu^2 R$

$= 4 - (0.1)^2 \times 150$

$= 2.5\ mm.$

2nd:

$h_0 = 30mm$ $h_f = 10mm$

$D = 600mm$ $\mu = 0.1$

$N = \frac{\Delta h_{require}}{\Delta h_{max}}$

$\Delta h = 30 - 10 = 20mm.$

$\Delta h_{max} = \mu^2 R = (0.1)^2 \times 300$

$= 3$

$N = \frac{20}{3} = 6.66 \simeq 7$

IES'01

$h_0 = 30m$ $h_f = 15m$ $D = 300$

$\mu \geq \tan\alpha$ $\mu = ?$

$\Delta h = h_0 - h_f = D(1-\cos\alpha)$

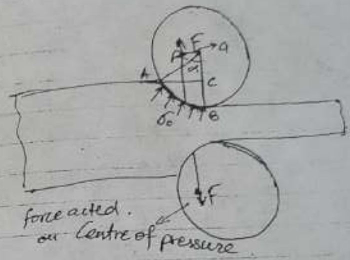
~~$\Rightarrow 15 = 300$~~

$\Delta h_{max} = \mu^2 R$

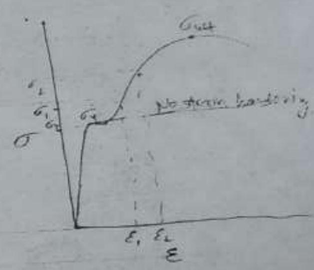
$30 - 15 = \mu^2 \times 150$

$\mu^2 = \sqrt{\frac{15}{150}} = \sqrt{\frac{1}{10}} = \frac{1}{3} = 0.33$

Force, Torque & Power in Rolling:-



force acted on Centre of pressure



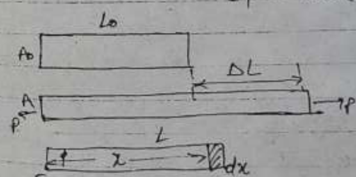
σ_0 = flow stress.

$= K\varepsilon_T^n$ with strain hardening

$\sigma_0 = \sigma_y$ without strain hardening.

True strain (ε_T)

$\varepsilon_T = \ln(1+\varepsilon) = \ln\frac{L}{L_0}$



Engineering strain or conventional strain $\varepsilon = \frac{\Delta L}{L_0}$

True strain (ε_T) = $\frac{\text{Elongation}}{\text{Instantaneous length}}$

$= \int_{L_0}^{L} \frac{dx}{x} = \ln\frac{L}{L_0}$

$= \ln(1+\varepsilon)$

$\frac{L}{L_0} = \frac{L_0 + \Delta L}{L_0}$

$= 1 + \frac{\Delta L}{L_0}$

$= 1 + \varepsilon$

[in tension positive value]

Rolling

Formula in Rolling:-

Reduction or Draft $(\Delta h) = h_0 - h_f$

Arc of contact $(AB) = R \cdot \alpha$ Rad.

$\alpha \rightarrow$ angle of bite

$BC = \frac{h_0}{2} - \frac{h_f}{2} = \frac{\Delta h}{2}$

$DC = R - \frac{\Delta h}{2}$

$\cos\alpha = \frac{DC}{DA} = \frac{R - \Delta h/2}{R}$

or $R\cos\alpha = R - \frac{\Delta h}{2}$

or, $\frac{\Delta h}{2} = R - R\cos\alpha$

or $\boxed{\Delta h = 2R(1-\cos\alpha)}$

or $\boxed{\Delta h = D(1-\cos\alpha)}$ VIMP

$h_0 = 16mm$ $h_f = 10mm$ $D = 400mm$

$\Delta h = h_0 - h_f = D(1-\cos\alpha)$

$16 - 10 = 400(1-\cos\alpha)$

$\cos\alpha = 1 - \frac{6}{400} = \frac{394}{400}$

$\alpha = 9.936°$ rad.

Ques[12] $D = 410mm$ $h_0 = 140mm$ Reduction $\Delta h_f = 8mm$, in 10% by

$\Delta h = 0.8$

$0.8 = 410(1-\cos\alpha)$

$\cos\alpha = 1 - \frac{0.8}{410}$

$\alpha = 3.579°$ $= 3.579 \times \frac{\pi}{180}$ rad.

Unaided entry (Rolling).

$\mu \geq \tan\alpha$.

$\mu = \tan\alpha$ [extreme condn]

Max. Reduction or Max. draft $\boxed{\Delta h_{max} = \mu^2 R}$.

$h_0 - h_{f\,min} = \mu^2 R$.

$\boxed{h_{f\,min} = h_0 - \mu^2 R}$

Min. possible thickness in a single pass.

No. of pass required $N = \frac{(\Delta h)_{required}}{(\Delta h)_{max}}$

$V_1 = 50$ m/min $\quad T_1 = 45$ min
$V_2 = 100$ m/min $\quad T_2 = 10$ min

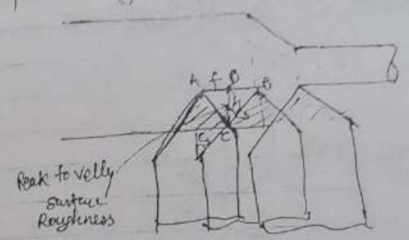
$V_1 T_1^n = C$
$V_2 T_2^n = C$
$V_1 T_1^n = V_2 T_2^n$
$50 \times 45^n = 100 \times 10^n$
$2 = (4.5)^n$
$\log 2 = n \log 4.5$
$n = 0.46$
$50 \times 45^{0.46} = C$
$C = 288$

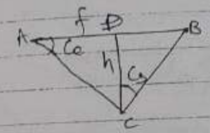
Max^n productivity for 2 mins $\qquad V_0 = ?$

$T_0 = T_c\left(\frac{1-n}{n}\right)$
$= 2\left(\frac{1-0.46}{0.46}\right)$
$= 2.34$ min.

$V_0 = \frac{C}{T_0^n} = \frac{288}{2.34^{0.46}} = 194$ m/min

Surface Roughness (In Turning Operation):





$\frac{AD}{h} = \cot C_e$
$\frac{DB}{h} = \tan C_s$

$AD + BD = f$
$h \cot C_e + h \tan C_s = f$

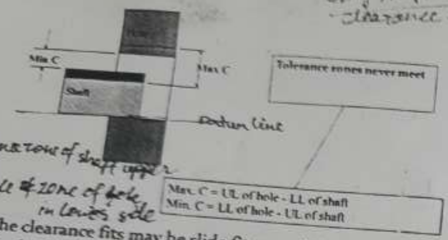
Roughness $\boxed{h = \frac{f}{\tan C_s + \cot C_e}}$ Ideal surface [zero nose Radius.

Ra or CLA (Center line average) $= \frac{h}{4}$
Roughness average

with nose Radius $\boxed{h = \frac{f^2}{8R}}$ & $\boxed{R_a = \frac{f^2}{18\sqrt{3}R}}$

Clearance Fits

UL of shaft < LL of hole
clearance is betw

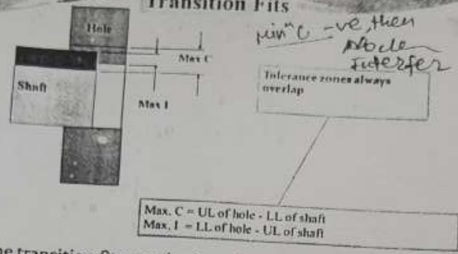


Tolerance zone of shaft upper
tolerance & zone of hole in lower side

The clearance fits may be slide fit, easy sliding fit, running fit, slack running fit and loose running fit.

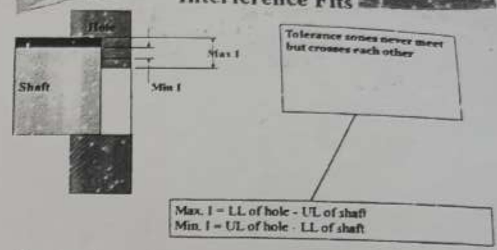
Transition Fits

min C -ve, then mode interfer



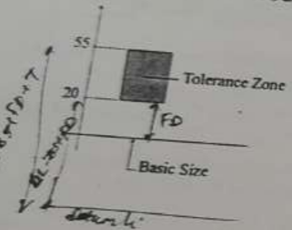
The transition fits may be ~~force~~ fit, tight fit and push fit.
Pressfit

Interference Fits



The interference fits may be shrink fit, heavy drive fit and light drive fit.

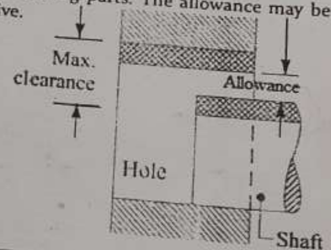
Tolerance Zone



- It is defined graphically by the magnitude of the tolerance and by its position in relation to the zero line.

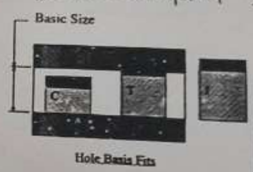
Allowance

- It is Minimum clearance or maximum interference. It is the *intentional* difference between the basic dimensions of the mating parts. The allowance may be positive or negative.



5. Basis of Fits - Hole Basis

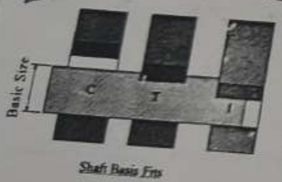
H → Hole basis System



- In this system, the basic diameter of the hole is constant while the shaft size varies according to the type of fit.

Legends:
Hole — C - Clearance
Shaft — T - Transition
Tolerance — I - Interference

Basis of Fits - Shaft Basis



- Here the hole size is varied to produce the required class of fit with a basic-size shaft.
- A series of drills and reamers is required for this system, therefore it tends to be costly.
- It may, however, be necessary to use it where different fits are required along a long shaft. For example, in the case of driving shafts where a single shaft may have to accommodate to a variety of accessories such as couplings, bearings, collars, etc., it is preferable to maintain a constant diameter for the permanent member, which is the shaft, and vary the bore of the accessories.

Legends:
Hole — C - Clearance
Shaft — T - Transition
Tolerance — I - Interference

Limits and Fits

- Limits and fits comprises 18 grades of fundamental tolerances for both shaft and hole, designated as IT01, IT0 and IT1 to IT16. These are called standard tolerances. (IS-919) But ISO 286 specify 20 grades upto IT18
- There are 25 (IS 919) and 28 (ISO 286) types of fundamental deviations.

Hole: A, B, C, CD, D, E, EF, F, FG, G, H, J JS, K, M, N, P, R, S, T, U, V, X, Y, Z, ZA, ZB, ZC.
Shaft : a, b, c, cd, d, e, ef, f, fg, g, h, j, js, k m, n, p, r, s, t, u, v, x, y, z, za, zb, zc.

- A unilateral hole basis system is recommended but if necessary a unilateral or bilateral shaft basis system may also be used

Tolerance Designation (ISO)

Tolerance on a shaft or a hole can also be calculated by using the formulas provided by ISO.

$$\boxed{T = K \times i}$$

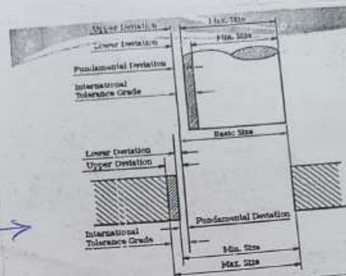
where,
T is the tolerance (in μm)
$i = 0.45\sqrt[3]{D} + 0.001D$ (unit tolerance, in μm)
Fundamental or standard
$D = \sqrt{D_1 D_2}$ (D_1 and D_2 are the nominal sizes marking the beginning and the end of a range of sizes, in mm)
Geometric mean Diameter
$K = 10(1.6)^{(IT_n - IT6)}$ [For IT6 to IT16]

Methodology

Limit, Tolerance & Fits

By S K Mondal

Orderly



Terminology

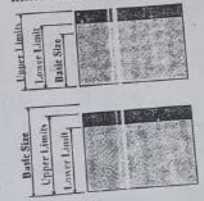
- **Nominal size:** Size of a part specified in the drawing. loose, without tolerances
- **Basic size:** Size of a part to which all limits of variation (i.e. tolerances) are applied.
- (of a product) **Actual size:** Actual measured dimension of the part. The difference between the basic size and the actual size should not exceed a certain limit, otherwise it will interfere with the interchangeability of the mating parts.

Terminology
Contd....

- **Limits of sizes:** There are two extreme permissible sizes for a dimension of the part. The largest permissible size for a dimension is called upper or high or maximum limit, whereas the smallest size is known as lower or minimum limit.
- **Tolerance** T = UL − LL
  - The difference between the upper limit and lower limit.
  - It is the maximum permissible variation in a dimension.
  - The tolerance may be unilateral or bilateral.

Terminology
Contd....

Unilateral Limits occurs when both maximum limit and minimum limit are either above or below the basic size.



e.g. $\varnothing 25^{+0.18}_{+0.10}$ Unilateral

Basic Size = 25.00 mm
Upper Limit = 25.18 mm
Lower Limit = 25.10 mm
Tolerance = 0.08 mm

+ve deviation

e.g. $\varnothing 25^{-0.10}_{-0.20}$

Basic Size = 25.00 mm
Upper Limit = 24.90 mm
Lower Limit = 24.80 mm
Tolerance = 0.10 mm

−ve deviation

Terminology
Contd....

For Unilateral Limits, a case may occur when one of the limits **coincides** with the basic size,

e.g. $\varnothing 25^{+0.20}_{0}$, $\varnothing 25^{0}_{-0.10}$ Unilateral limits

Bilateral Limits occur when the maximum limit is above and the minimum limit is below the basic size.

e.g. $\varnothing 25 \pm 0.04$ Bilateral limits
Basic Size = 25.00 mm
Upper Limit = 25.04 mm
Lower Limit = 24.96 mm
Tolerance = 0.08 mm

mostly applied, more acceptable

Terminology
Contd....

zero deviation

- **Zero line:** A straight line corresponding to the basic size. The deviations are measured from this line.
- **Deviation:** Is the algebraic difference between a size (actual, max. etc.) and the corresponding basic size.
- **Actual deviation:** Is the algebraic difference between an actual size and the corresponding basic size.
- **Upper deviation:** Is the algebraic difference between the maximum size and the basic size.

Terminology
Contd....

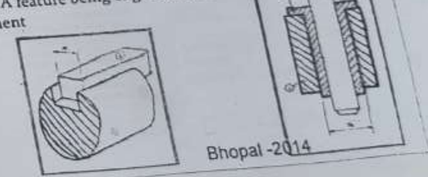
- **Lower deviation:** Is the algebraic difference between the minimum size and the basic size.
- **Mean deviation:** Is the arithmetical mean of upper and lower deviations.
- **Fundamental deviation:** This is the deviation, either the upper or the lower deviation, which is nearest one to zero line for either a hole or shaft.

Fit

Fits: (assembly condition between "Hole" & "Shaft")

Hole – A feature engulfing a component

Shaft – A feature being engulfed by a component



Projected length (L_p) = ... $R\sin\theta = \sqrt{R\Delta h}$ mm

Projected Area(A_p) = $L_p \times b$, mm^2

Roll Separating force(F) = $\sigma_0 L_p b$, N

Arm length(a) = $0.5\, L_p$ for hot rolling
= $0.45\, L_p$ for cold rolling } mm

Torque per roller $(T) = \dfrac{F \times a}{1000}$, N.m

Total Power for two roller (P) = $2\,T.\omega$, W

$h_0 = 20mm$ $b = 100mm$ $h_f = 18$ $R = 250mm$
$N = 10\,rpm$ $\sigma_0 = 300MPa$ $P = ?$

$F = \sigma_0 L_p b = 300 \times 22.36 \times 100 = 670{,}820.39\,N$

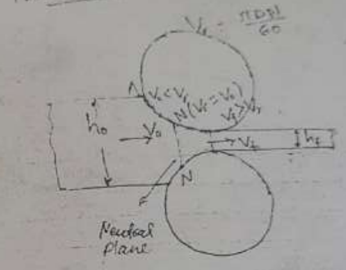
$L_p = \sqrt{R\Delta h} = \sqrt{250 \times (20-18)} = 22.36\,mm$

$P = 2.F.\omega = 2 \times F \times a \times \omega / 1000$
$= \dfrac{2 \times 670820.39 \times 0.45 \times 2\pi \times 10}{1000 \times 1000 \times 60}$ kW

$= 15.64\,kW$

In rolling, force velocity formula is not applicable because dir^n of force & dir of velocity are not same

Neutral Plane :-



Backward slip = $\dfrac{V_r - V_0}{V_r} \times 100\%$

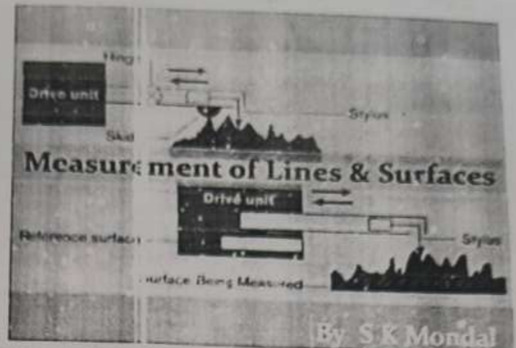
Forward slip = $\dfrac{V_f - V_r}{V_r} \times 100\%$

$D_1 = 200\,mm$ $D_2 = 400\,mm$ $h = 60\,mm$
$\varepsilon_T = ?$

Ground + Grinding
Hot forged on hammer
Heat treatment
Polishing

Taylor's Principle

This principle states that the GO gauge should always be so designed that it will cover the maximum metal condition (MMC) of as many dimensions as possible in the same limit gauges, whereas a NOT GO gauges to cover the minimum metal condition of one dimension only.



Linear measurements

Some of the instruments used for the linear measurements are:

- Rules [scales] [accuracy low] only estimate
- Vernier
- Micrometer → In workshop
- Height gauge
- Bore gauge
- Dial indicator
- Slip gauges or gauge blocks [Better accuracy] [Time Consuming]

Vernier Caliper

- A vernier scale is an auxiliary scale that slides along the main scale.
- The vernier scale is that a certain number n of divisions on the vernier scale is equal in length to a different number (usually one less) of main-scale divisions.

$$nV = (n-1)S$$

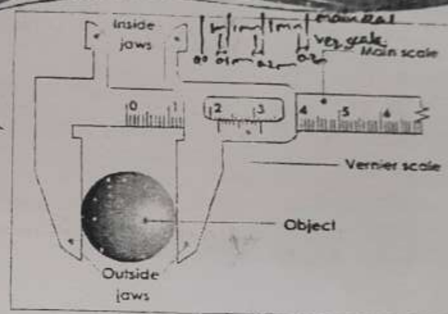
where n = number of divisions on the vernier scale

V = The length of one division on the vernier scale

and S = Length of the smallest main-scale division

- Least count is applied to the smallest value that can be read directly by use of a vernier scale.
- Least count $= S - V = \frac{1}{n}S$

Least count = 0.1mm



Vernier Caliper

Metric Micrometer

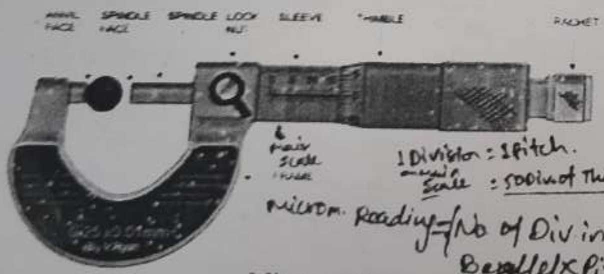
- A *micrometer* allows a measurement of the size of a body. It is one of the most accurate mechanical devices in common use.
- It consists a main scale and a thimble

Method of Measurement

Step-I: Find the whole number of mm in the barrel

Step-I: Find the reading of barrel and multiply by 0.01

Step-III: Add the value in Step-I and Step-II



Micrometer

1 Division = 1 Pitch.

main scale : 500 div of Thimble

Microm. Reading = (No of Div in the Barrel × Pitch) + (Thimble Reading × Pitch) (No of Divs. in the Thimble)

- **Bore Gauge:** used for measuring bores of different sizes ranging from small-to-large sizes.
- Provided with various extension arms that can be added for different sizes.

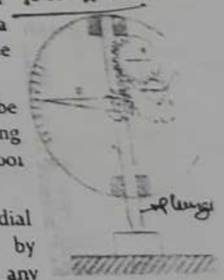
Displacement sensor

- **Dial indicator:** Converts a linear displacement into a radial movement to measure over a small range of movement for the plunger.

Mostly used

1 div : 1 μm

- The typical least count that can be obtained with suitable gearing dial indicators is 0.01 mm to 0.001 mm.
- It is possible to use the dial indicator as a comparator by mounting it on a stand at any suitable height.



Principle of a dial indicator

Selective Assembly

- All the parts (hole & shaft) produced are measured and graded into a range of dimensions within the tolerance groups. [transportation + Handling]
- Reduces the cost of production

$100^{\pm 0.25}$ by any m/c

$100^{\pm 0.025}$ on CNC

[Improve cost of product]

Limit Gauges

Selection By → Attributes → Variables

- **Plug gauge:** used to check the holes. The GO plug gauge is the size of the low limit of the hole while the NOT GO plug gauge corresponds to the high limit of the hole. → Attribute
- **Snap, Gap or Ring gauge:** used for gauging the shaft and male components. The Go snap gauge is of a size corresponding to the high (maximum) limit of the shaft, while the NOT GO gauge corresponds to the low (minimum limit).

for bolt — for shaft

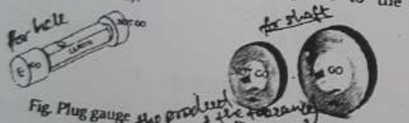
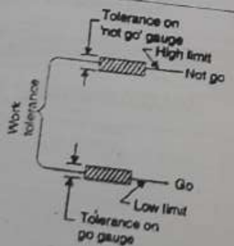


Fig. Plug gauge — Fig. Ring and snap gauges

[To check the product under and tolerance]

- **Bilateral system:** in this system, the GO and NO GO gauge tolerance zones are bisected by the high and low limits of the work tolerance zone.



Taking example as above:

$\therefore$ Dimension of 'GO' Plug gauge $= 24.98^{+0.002}_{-0.002}$ mm

Dimension of 'NOT GO' Plug gauge $= 25.02^{+0.002}_{-0.002}$ mm

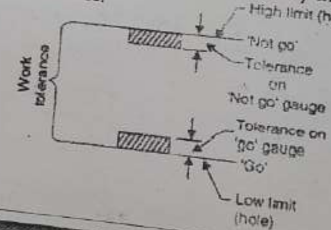
For IES, GATE, PSUs

Interchangeability

- Term employed for the mass production of identical items within the prescribed limits of sizes.
- If the variation of items are within certain limits, all parts of equivalent size will be equally fit for operating in machines and mechanisms and the mating parts will give the required fitting.
- This facilitates to select at random from a large number of parts for an assembly and results in a considerable saving in the cost of production.

Allocation of manufacturing tolerances

- **Unilateral system:** gauge tolerance zone lies entirely within the work tolerance zone.
- work tolerance zone becomes smaller by the sum of the gauge tolerance.



- **Wear allowance:** GO gauges which constantly rub against the surface of the parts in the inspection are subjected to wear and loose their initial size.
- The size of go plug gauge is reduced while that of go snap gauge increases.
- To increase service life of gauges wear allowance is added to the go gauge in the direction opposite to wear. Wear allowance is usually taken as 5% of the work tolerance.
- Wear allowance is applied to a nominal diameter before gauge tolerance is applied.

Tolerance Sink

Tolerance sink

- A design engineer keeps one section of the part blank (without tolerance) so that production engineer can dump all the tolerances on that section which becomes most inaccurate dimension of the part.
- Position of sink can be changing the reference point.
- Tolerance for the sink is the cumulative sum of all the tolerances and only like minded tolerances can be added i.e. either equally bilateral or equally unilateral.

Example

Size of the hole to be checked 25 ± 0.02 mm

Here, Higher limit of hole = 25.02 mm

Lower limit of hole = 24.98 mm

Work tolerance = 0.04 mm

$\therefore$ Gauge tolerance = 10% of work tolerance = 0.004 mm

$\therefore$ Dimension of 'GO' Plug gauge $= 24.98^{+0.004}_{-0.000}$ mm

Dimension of 'NOT GO' Plug gauge $= 25.02^{+0.000}_{-0.004}$ mm

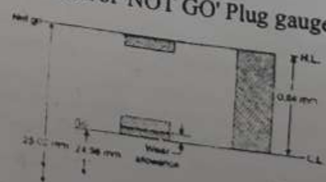
- Taking example of above:

$\therefore$ *Wear Allowance* = 5% of work tolerance = 0.002 mm

Nominal size of GO plug gauge = 24.98 + 0.002 mm

$\therefore$ Dimension of 'GO' Plug gauge $= 24.982^{+0.004}_{-0.000}$ mm

Dimension of 'NOT GO' Plug gauge $= 25.02^{+0.000}_{-0.004}$ mm



Diameter Steps

Above (mm)	Upto and including (mm)
-	3
3	6
6	10
10	18
18	30
30	50
50	80
80	120
120	180
180	250
250	315
315	400
400	500

Value of the Tolerance

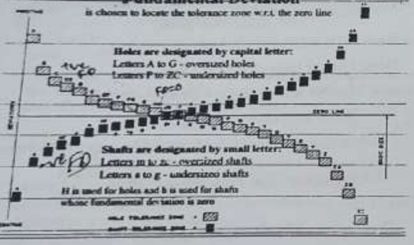
IT01 $0.3 + 0.008D$	IT0 $0.5 + 0.012D$	IT1 $0.8 + 0.02D = a$	IT2 ar, $r = 10^{1/5}$
IT3 ar^2	IT4 ar^3	IT5 $ar^4 = 7i$	IT6 $10(1.6)^{(IT_n - IT6)} = 10i$
IT7 $10(1.6)^{(IT_n - IT6)} = 16i$	IT8 $10(1.6)^{(IT_n - IT6)} = 25i$	IT9 $10(1.6)^{(IT_n - IT6)} = 40i$	IT10 $10(1.6)^{(IT_n - IT6)} = 64i$
IT11 $10(1.6)^{(IT_n - IT6)} = 100i$	IT12 $10(1.6)^{(IT_n - IT6)} = 160i$	IT13 $10(1.6)^{(IT_n - IT6)} = 250i$	IT14 $10(1.6)^{(IT_n - IT6)} = 400i$
IT15 $10(1.6)^{(IT_n - IT6)} = 640i$	IT16 $10(1.6)^{(IT_n - IT6)} = 1000i$		

$T = 16i$

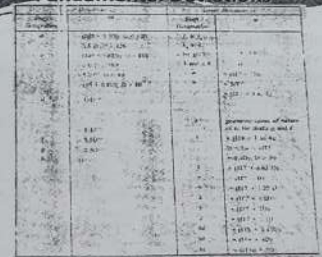
Grades of Tolerance

- It is an indication of the level of accuracy.
- IT01 to IT4 - For production of gauges, plug gauges, measuring instruments
- IT5 to IT 7 - For fits in precision engineering applications
- IT8 to IT11 - For General Engineering
- IT12 to IT14 - For Sheet metal working or press working
- IT15 to IT16 - For processes like casting, general cutting work

Fundamental Deviation



Fundamental Deviations



Calculation for Upper and Lower Deviation

- **For Shaft**

$ei = es - IT$ lower deviation

$es = ei + IT$

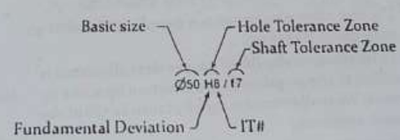
- **For Hole**

$EI = ES - IT$

$ES = EI + IT$

es = upper deviation of shaft
ei = lower deviation of shaft
ES = upper deviation of hole
EI= lower deviation of hole

- For hole, H stands for a dimension whose lower deviation refers to the basic size. The hole H for which the lower deviation is zero is called a basic hole.
- Similarly, for shafts, h stands for a dimension whose upper deviation refers to the basic size. The shaft h for which the upper deviation is zero is called a basic shaft.
- A fit is designated by its basic size followed by symbols representing the limits of each of its two components, the hole being quoted first.
- For example, 100 H6/g5 means basic size is 100 mm and the tolerance grade for the hole is 6 and for the shaft is 5.



Recommended Selection of Fits

TYPE OF FIT	SHAFT AND TOLERANCE	HOLE AND TOLERANCE			
		H7	H8	H9	H11
CLEARANCE	c11				
	d10				
	e9				
	f7				
	g6				
	h6				
TRANSITION	k6				
	n5				
INTERFERENCE	p6				
	s6				

Angular Measurement

This involves the measurement of angles of tapers and similar surfaces. The most common angular measuring tools are:

- Bevel protractor
- Sine bar

deg.

12 lines each side

1 of 5 min

Bevel Protractor

- Is part of the machinist's combination square.
- The flat base of the protractor helps in setting it firmly on the workpiece and then by rotating the rule, it is possible to measure the angle. It will typically have a discrimination of one degree.

12 div. of VS = 23 div of MS

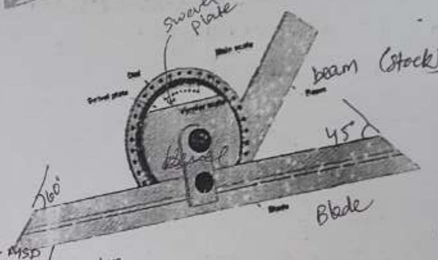
1 MSD = 1° $1^\circ = 60'$ 1 VSD = $\frac{23}{12}$ MSD

1 VSD = 5' $1' = 60''$ 1 MSD = 1°

1 VSD = $\frac{23}{12}$°

LC = 2 MSD − 1 VSD

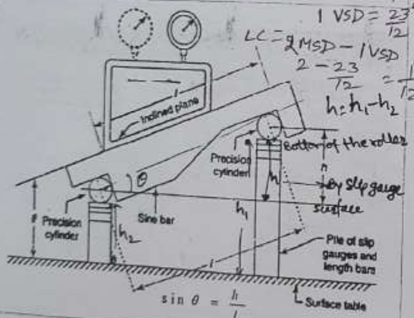
$2 - \frac{23}{12} = \frac{1}{12}^\circ \rightarrow \frac{60'}{12} = 5'$



A Bevel Protractor

Sine Bar

- When a reference for a non-square angle is required, a sine bar can be used.
- Basically a sine bar is a bar of known length. When gauge blocks are placed under one end, the sine bar will tilt to a specific angle.
- Knowing the height differential of the two rollers in alignment with the workpiece, the angle can be calculated using the sine formula.
- A sine bar is specified by the distance between the centre of the two rollers, i.e. 100 mm, 200 mm, & 300 mm. the various part of sine bar are hardened before grinding & lapping.

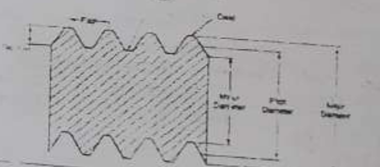


$h = h_1 - h_2$

$\sin\theta = \frac{h}{l}$

Thread Measurements

- Threads are normally specified by the major diameter.
- Though there are a large variety of threads used in engineering, the most common thread encountered is the metric V-thread shown in Fig.

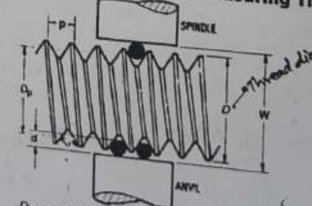


- The parameters that are normally measured are:
- Major diameter
  - Micrometer
- Pitch diameter
  - Screw thread micrometer
  - Wire method (pitch dia of the thread)
- Pitch
  - Screw pitch gauge
  - Pitch measuring machine
- Thread form
  - Optical projector

Three-Wire Method

- Three wires of equal diameter placed in thread, two on one side and one on other side
- Standard micrometer used to measure distance over wires (M)
- Different sizes and pitches of threads require different sizes of wires

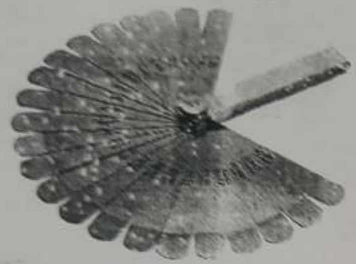
The Three-Wire Method of Measuring Threads



D_p = pitch diameter or Effective diameter

p = pitch of thread, and α

Feeler Gauge

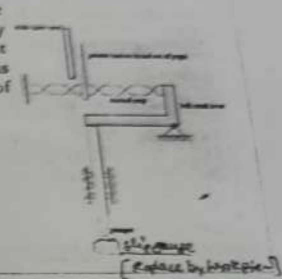


Limit Gauges

Gauge	For Measuring
Snap Gauge	External Dimensions
Plug Gauge	Internal Dimensions
Taper Plug Gauge	Taper hole
Ring Gauge	External Diameter
Gap Gauge	Gaps and Grooves
Radius Gauge	Gauging radius
Thread pitch Gauge	External Thread

Mechanical Comparators

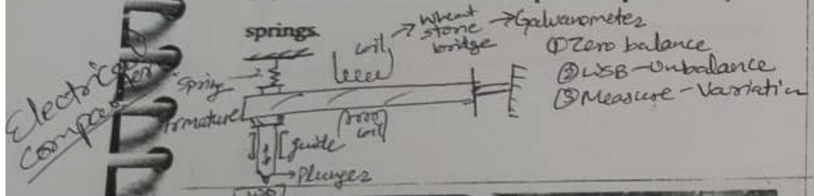
- The Mikrokator principle greatly magnifies any deviation in size so that even small deviations produce large deflections of the pointer over the scale.

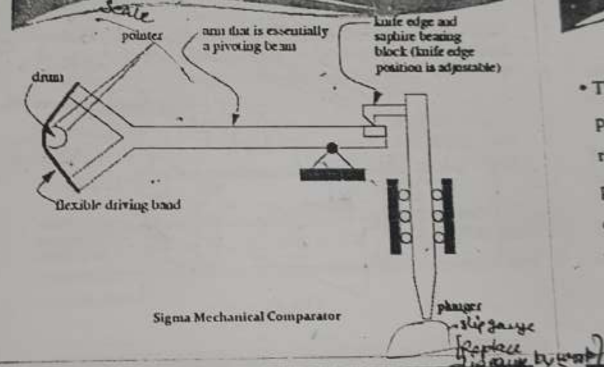


Sigma Mechanical Comparator

The Sigma Mechanical Comparator uses a partially wrapped band wrapped about a driving drum to turn a pointer needle. The assembly provides a frictionless movement with a resistant pressure provided by the springs.

Electrical comparator

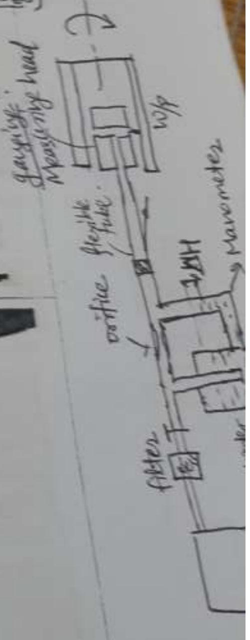




Sigma Mechanical Comparator

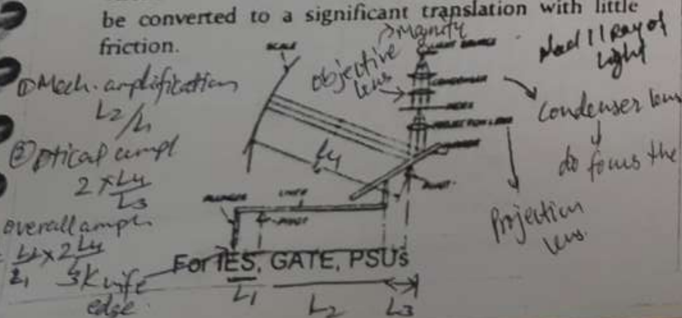
Mechanical Comparators (Recovery Base)

- The Eden-Rolt Reed system uses a pointer attached to the end of two reeds. One reed is pushed by a plunger, while the other is fixed. As one reed moves relative to the other, the pointer that they are commonly attached to will deflect.



Optical Comparators

- These devices use a plunger to rotate a mirror. A light beam is reflected off that mirror, and simply by the virtue of distance, the small rotation of the mirror can be converted to a significant translation with little friction.

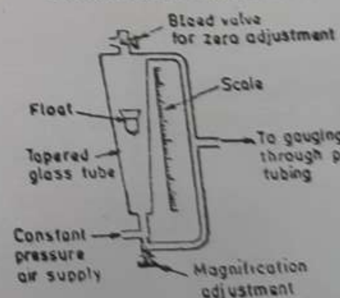


For IES, GATE, PSUs

Pneumatic Comparators

- Flow type:
  - The float height is essentially proportional to the air that escapes from the gauge head
  - Master gauges are used to find calibration points on the scales
  - The input pressure is regulated to allow magnification adjustment

Pneumatic Comparators



Applications of dial indicator include:

- centering workpieces to machine tool spindles
- offsetting lathe tail stocks
- aligning a vice on a milling machine
- checking dimensions

Slip Gauges or Gauge blocks

- These are small blocks of alloy steel.
- Used in the manufacturing shops as length standards.
- Not to be used for regular and continuous measurement.
- Rectangular blocks with thickness representing the dimension of the block. The cross-section of the block is usually 32 mm x 9 mm.
- Are hardened and finished to size. The measuring surfaces of the gauge blocks are finished to a very high degree of finish, flatness and accuracy.

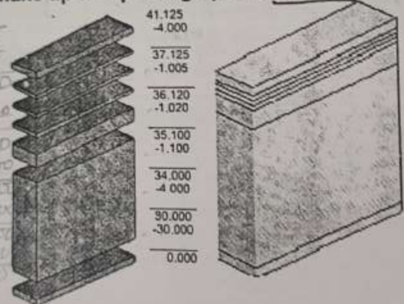
- Come in sets with different number of pieces in a given set to suit the requirements of measurements.
- A typical set consisting of 88 pieces for metric units is shown in.
- To build any given dimension, it is necessary to identify a set of blocks, which are to be put together.
- Number of blocks used should always be the smallest.
- Generally the top and bottom Slip Gauges in the pile are 2 mm wear gauges. This is so that they will be the only ones that will wear down, and it is much cheaper to replace two gauges than a whole set.

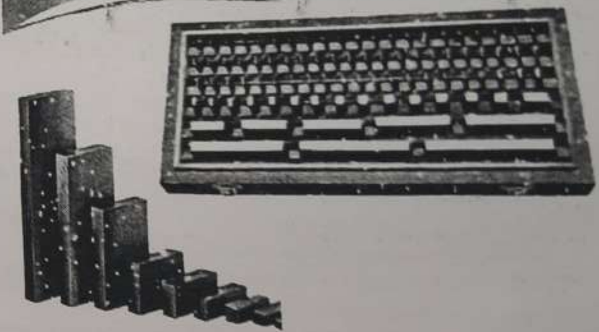
To make up a Slip Gauge pile to 41.125 mm

- A Slip Gauge pile is set up with the use of simple maths.
- Decide what height you want to set up, in this case 41.125mm.
- Take away the thickness of the two wear gauges, and then use the gauges in the set to remove each place of decimal in turn, starting with the lowest.

To make up a Slip Gauge pile to 41.125 mm

41.125
-1.005
40.120
-1.020
39.100
-1.100
38.000
-8.000
30.000
-30.000
0.000





A Metric slip gauge set (88 Pieces)

Slip gauges size or range, mm	Increment, mm	Number of Pieces
1.005	-	1
1.001 to 1.009	0.001	9
1.010 to 1.490	0.010	49
0.500 to 9.500	0.500	19
10 to 100	10.000	10

Comparators (compare Dimension)

- Comparator is another form of linear measuring method, which is quick and more convenient for checking large number of identical dimensions.
- During the measurement, a comparator is able to give the deviation of the dimension from the set dimension.
- Cannot measure absolute dimension but can only compare two dimensions.
- Highly reliable.
- To magnify the deviation, a number of principles are used such as mechanical, optical, pneumatic and electrical.

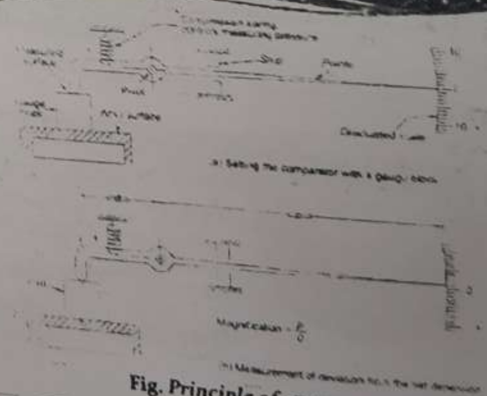


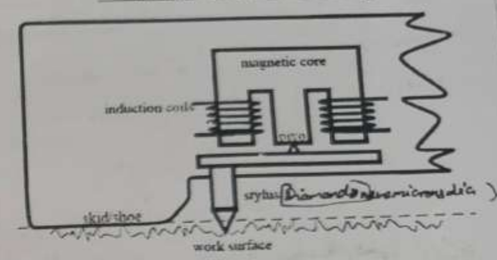
Fig. Principle of a Bhopal-2014

Observation Methods

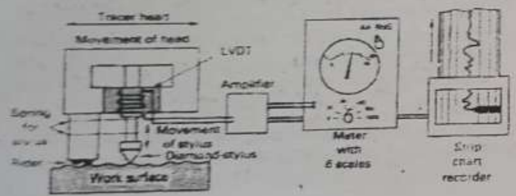
- Human perception is highly relative.
- To give the human tester a reference for what they are touching, commercial sets of standards are available.
- Comparison should be made against matched identical processes.
- One method of note is the finger nail assessment of roughness and touch method.

Stylus Equipment

- uses a stylus that tracks small changes in surface height, and a skid that follows large changes in surface height.
- The relative motion between the skid and the stylus is measured with a magnetic circuit and induction coils.
- One example of this is the Brown & Sharpe Surfcom unit.



Schematic of stylus profile device for measuring surface roughness and surface profile with two readout devices shown: a meter for AA or rms values and a strip-chart recorder for surface profile.



Profilometer

- measuring instrument used to measure a surface's profile, in order to quantify its roughness.
- Vertical resolution is usually in the nanometre level, though lateral resolution is usually poorer.

Contact profilometers

- A diamond stylus is moved vertically in contact with a sample and then moved laterally across the sample for a specified distance and specified contact force.
- A profilometer can measure small surface variations in vertical stylus displacement as a function of position.
- The radius of diamond stylus ranges from 20 nanometres to 25 μm.

Non-contact Profilometers

- An optical profilometer is a non-contact method for providing much of the same information as a stylus based profilometer.
- There are many different techniques which are currently being employed, such as laser triangulation (triangulation sensor), confocal microscopy and digital holography.

Advantages of optical Profilometers

costly

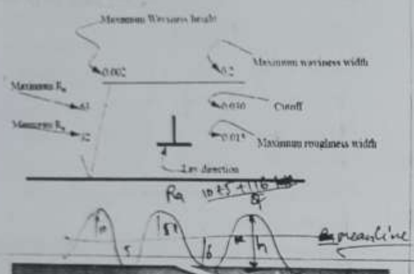
- Because the non-contact profilometer does not touch the surface the scan speeds are dictated by the light reflected from the surface and the speed of the acquisition electronics.
- Optical profilometers do not touch the surface and therefore cannot be damaged by surface wear or careless operators.

Optical Flats

of quartz

- Optical-grade glass structures or clear fused quartz lapped and polished to be extremely flat on one or both sides.
- Used with a monochromatic light to determine the flatness of other optical surfaces by interference.
- When a flat surface of another optic is placed on the optical flat, interference fringes are seen due to interference in the tiny gap between the two surfaces.
- The spacing between the fringes is smaller where the gap is changing more rapidly, indicating a departure from flatness in one of the two surfaces, in a similar way to the contour lines on a map.

Representation of Surface Roughness



Roughness Ra	Roughness Grade Number	Roughness Symbol
	N12	
50/2=25	N11	▽ Grindi.
25/2=11	N10	
12.5/2=6.25	N9	
6.3/2=3.15=3.2	N8	▽▽ finishing
3.2/2	N7	
1.6/2	N6	
0.8	N5	▽▽▽ Superfinishing
0.4	N4	
0.2	N3	
0.1	N2	▽▽▽▽
0.05/2=0.025	N1	
0.025		

- **Waviness height** - the distance from a peak to a valley.
- **Waviness width** - the distance between peaks or valleys
- **Roughness width cutoff** - a value greater than the maximum roughness width that is the largest separation of surface irregularities included in the measurements. Typical values are (0.003", 0.010", 0.030", 0.100", 0.300")
- **Lay** - the direction the roughness pattern should follow
  - Stylus travel is perpendicular to the lay specified.

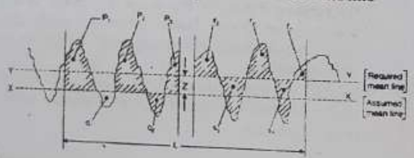
Evaluation of Surface Roughness

1. Centre line average (CLA) or arithmetic mean deviation denoted as R_a.
2. Root mean square value (R_g) : rms value
3. Maximum peak to valley roughness (h_{max})
4. The average of the five highest peak and five deepst valleys in the sample.
5. The average or leveling depth of the profile.

(flatten the p... By roller burnishing)

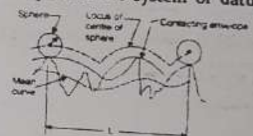
Determination of Mean Line

- **M-System:** After plotting the characteristic of any surface a horizontal line is drawn by joining two points. This line is shifts up and down in such a way that 50% area is above the line and 50% area is below the line



Determination of Mean Line

- **E-System: (Envelop System)** A sphere of 25 mm diameter is rolled over the surface and the locus of its centre is being traced out called envelope. This envelope is shifted in downward direction till the area above the line is equal to the area below the line. This is called mean envelope and the system of datum is called E-system.



Arithmetical Average:

- Measured for a specified area and the figures are added together and the total is then divided by the number of measurements taken to obtain the mean or arithmetical average (AA).
- It is also sometimes called the centre line average or CLA value. This in equation form is given by

$$R_a = \frac{1}{L}\int_0^L |y(x)|\,dx \cong \frac{1}{N}\sum |y_i|$$

- The other parameter that is used sometimes is the root mean square value of the deviation in place of the arithmetic average , This in expression form is

$$R_{RMS} = \sqrt{\frac{1}{N}\sum y_i^2}$$

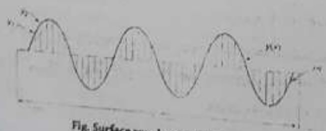


Fig. Surface roughness parameters

Methods of measuring Surface Roughness

There are a number of useful techniques for measuring surface roughness:

- **Observation and touch** (Relative) - the human finger is very perceptive to surface roughness
- **stylus based equipment** - very common
- **Interferometry** - uses light wave interference patterns (discussed later)

Distance W over the outer edge

$$W = D_r + d\left(1 + \operatorname{cosec}\frac{\alpha}{2}\right) - \frac{p}{2}\cot\frac{\alpha}{2}$$

For ISO metric thread, $\alpha = 60$ and $D_r = D - 0.6496p$

$$W = D + 3d - 1.5156p$$

Best wire size — IMP

$$d = \frac{p}{2}\sec\frac{\alpha}{2}$$

Effective dia → Pitch dia.

For ISO metric thread, $\alpha = 60$

$$d = 0.5774p$$

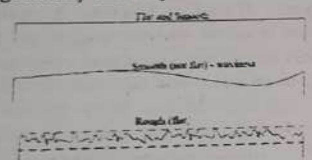
Measurement of Surfaces

→ Roughness
→ Profile (Roughness + waviness)

Surfaces

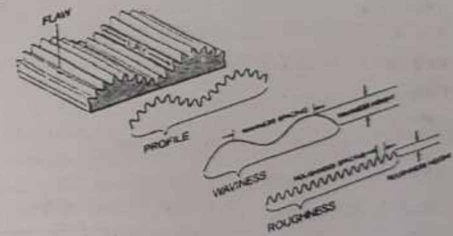
- No surface is perfectly smooth, but the better the surface quality, the longer a product generally lasts, and the better is performs.
- Surface texture can be difficult to analyse quantitatively.
- Two surfaces may be entirely different, yet still provide the same CLA (Ra) value.

- Surface geometry can be quantified a few different ways.



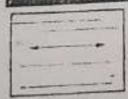
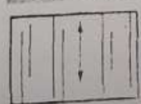
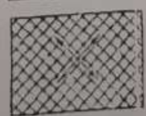
- Real surfaces are rarely so flat, or smooth, but most commonly a combination of the two.

- **Roughness height:** is the parameter with which generally the surface finish is indicated. It is specified either as arithmetic average value or the root mean square value.
- **Roughness width:** is the distance parallel to the nominal part surface within which the peaks and valleys, which constitutes the predominant pattern of the roughness.
- **Roughness width cut-off:** is the maximum width of the surface that is included in the calculation of the roughness height.

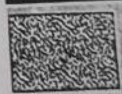


- **Waviness:** refers to those surface irregularities that have a greater spacing than that of roughness width.
  - Determined by the height of the waviness and its width.
  - The greater the width, the smoother is the surface and thus is more desirable.
- **Lay direction:** is the direction of the predominant surface pattern produced on the workpiece by the tool marks.
- **Flaw:** are surface irregularities that are present which are random and therefore will not be considered.

Lay

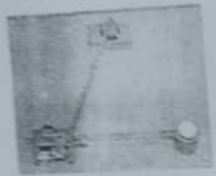
Diagram	Symbol	Description
	√=	**Parallel lay:** Lay parallel to the Surface. Surface is produced by shaping, planning etc.
	√⊥	**Perpendicular lay:** Lay perpendicular to the Surface. Surface is produced by shaping and planning
	√X	**Crossed lay:** Lay angular in both directions. Surface is produced by knurling, honing.

Lay Contd..

Diagram	Symbol	Description
	√M	**Multidirectional lay:** Lay multidirectional. Surface is produced by grinding, lapping, super finishing.
	√C	**Circular lay:** Approximately circular relative to the center. Surface is produced by facing.
	√R	**Radial lay:** Approximately radial relative to the center of the nominal surface.

Planimeter

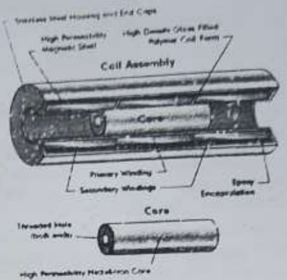
- A device used for measuring the area of any plane surface by tracing the boundary of the area.



LVDT

- Acronym for Linear Variable Differential Transformer, a common type of electromechanical transducer that can convert the rectilinear motion of an object to which it is coupled mechanically into a corresponding electrical signal.
- LVDT linear position sensors are readily available that can measure movements as small as a few millionths of an inch up to several inches, but are also capable of measuring positions up to ±20 inches (±0.5 m).
- A **rotary variable differential transformer (RVDT)** is a type of electrical transformer used for measuring angular displacement.

LVDT



Tool Maker's Microscope

An essential part of engineering inspection, measurement and calibration in metrology labs. Hence is used to the following:

- Examination of form tools, plate and template gauges, punches and dies, annular grooved and threaded hobs etc.
- Measurement of glass graticules and other surface marked parts.
- Elements of external thread forms of screw plug gauges, taps, worms and similar components.
- Shallow bores and recesses.

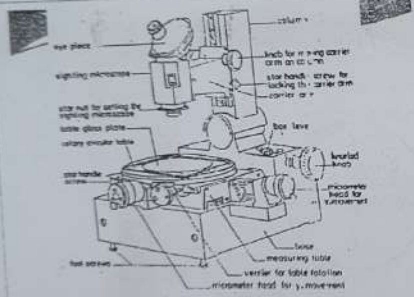


Fig 1 Tool Maker's Microscope

Telescopic Gauges

- Used to measure a bore's size, by transferring the internal dimension to a remote measuring tool.
- They are a direct equivalent of inside callipers and require the operator to develop the correct feel to obtain repeatable results.

Coordinate Measuring Machine (CMM)

- An instrument that locates point coordinates on three dimensional structures mainly used for quality control applications.
- The highly sensitive machine measures parts down to the fraction of an inch.
- Specifically, a CMM contains many highly sensitive air bearings on which the measuring arm floats.

Advantages,

- can automate inspection process
- less prone to careless errors
- allows direct feedback into computer system

Disadvantages,

- Costly
- fixturing is critical
- requires a very good tolerance model

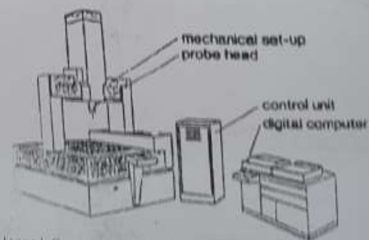


Figure 1. Components of a CMM

Autocollimator

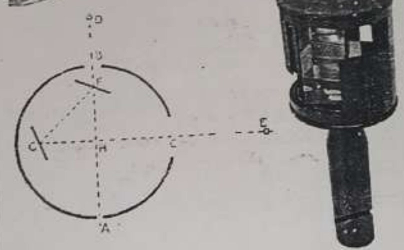
- An optical instrument for non-contact measurement of angles.
- Used to align components and measure deflections in optical or mechanical systems.
- An autocollimator works by projecting an image onto a target mirror, and measuring the deflection of the returned image against a scale, either visually or by means of an electronic detector.
- A visual autocollimator can measure angles as small as 0.5 arcsecond, while an electronic autocollimator can be up to 100 times more accurate.

- Visual autocollimators are used for lining up laser rod ends and checking the face parallelism of optical windows and wedges.
- Electronic and digital autocollimators are used as angle measurement standards, for monitoring angular movement over long periods of time and for checking angular position repeatability in mechanical systems.
- Servo autocollimators are specialized compact forms of electronic autocollimators that are used in high speed servo feedback loops for stable platform applications.

Autocollimator

Optical Square

- An Optical square consists of a small cylindrical metal box, about 5 cm in diameter and 12.5 cm deep, in which two mirrors are placed at an angle of 45° to each other and at right angles to the plane of the instrument.
- One mirror(horizon glass) is half silvered and other(index glass) is wholly silvered.
- The optical square belongs to a reflecting instruments which measure angles by reflection. Angle between the first incident ray and the last reflected ray is 90°
- Used to find out the foot of the perpendicular from a given point to a line.
- Used to set out right angles at a given point on a line in the field.
- Two mirrors may be replaced by two prisms.

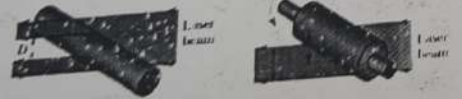


An Optical Square

Laser Scanning Micrometer

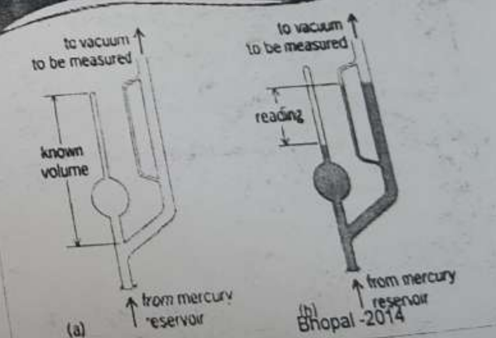
- The LSM features a high scanning rate which allows inspection of small workpiece even if they are fragile, at a high temperature, in motion or vibrating.
- Applications :
  - Measurement of outer dia. And roundness of cylinder,
  - Measurement of thickness of film and sheets,
  - Measurement of spacing if IC chips,
  - Measurement of forms,
  - Measurement of gap between rollers.

(a) Direct measurement of diameter D (b) Runout of shaft in rotation

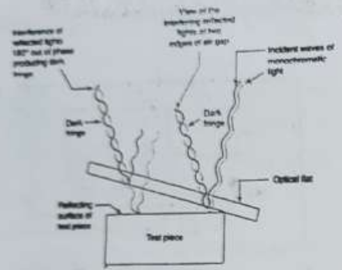


McLeod gauge

- Used to measure vacuum by application of the principle of Boyle's law.
- Works on the principle, "Compression of known volume of low pressure gas to higher pressure and measuring resulting volume & pressure, one can calculate initial pressure using Boyle's Law equation."
- Pressure of gases containing vapours cannot normally measured with a McLeod gauge, for the reason that compression will cause condensation .
- A pressure from 0.01 micron to 50 mm Hg can be measured. Generally McLeod gauge is used for calibration purpose.



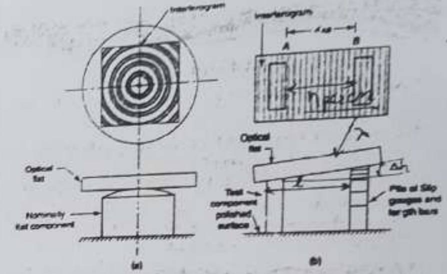
Bhopal -2014



- When the fringes are perfectly straight and same fringe width for dark and bright band we conclude that the surface is perfectly flat.
- For convex surface the fringes curve around the point of contact.
- For concave surface the fringes curve away from the point of contact.

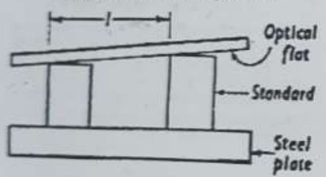
The distance of air gap between two successive fringes is given by $= \frac{\lambda}{2}$

Distance of air gap of interference fringe of n^{th} order is $= \frac{n\lambda}{2}$



(a) Interferogram of a nominally flat cylindrical component as seen on a circular flat. (b) Interferogram of a test set up of height difference between a component (top side polished) and slip gauge pile (polished slip gauges).

Optical flat as a comparator



$$\Delta h = \frac{n\lambda l}{2}$$ in cm

Where l = separation of edges

n = number of fringes / cm

Δh = The difference of height between gauges

λ = wevlength of monochomatic light

Talysurf

- It is based upon measuring the generated noise due to dry friction of a metallic blade which travels over the surface under consideration.
- If the frictional force is made small enough to excite the blade, and not the entire system, then the noise will be proportional to surface roughness, and independent of the measured specimen size and material.
- The specimen surface roughness was measured by a widely used commercial instrument (Talysurf 10), and the prototype transducer.

Clinometer

- An optical device for measuring elevation angles above horizontal.
- Compass clinometers are fundamentally just magnetic compasses held with their plane vertical so that a plummet or its equivalent can point to the elevation of the sight line.
- The clinometer can read easily and accurately angles of elevation that would be very difficult to measure in any other simple and inexpensive way.
- A fairly common use of a clinometer is to measure the height of trees.

Clinometer

Q. In drilling oprⁿ – Tool life ↓ from 20 min to 5 min due to ↑ in drill speed from 200 rpm to 400 rpm. What will be tool life of that drill under same condⁿ if drill speed is 300 rpm.

Ans. $N_1 T_1^n = N_2 T_2^n \Rightarrow 200(20)^n = 400(5)^n \Rightarrow 4^n = 2 \Rightarrow n = 0.5$

$N_1 T_1^n = N_3 T_3^n \Rightarrow T_3 = \left(\frac{200}{300}\right)^2 \times 20 = 8.89$ min

Q. An HSS tool is used for turning oprⁿ. The tool life is 1 hr when turning is carried out at 30 m/min, the tool life will be reduced to 2 min if the cutting speed is doubled. Find the suitable speed in RPM for turning 300 mm diameter so that tool life is 30 min.

As. $V_1 = 30$ m/min $\rightarrow T_1 = 1$ hr $= 60$ min
$V_2 = 2V_1 = 60$ m/min $\rightarrow T_2 = 2$ min

$V_1 T_1^n = V_2 T_2^n \Rightarrow \log V_1 + n \log T_1 = \log V_2 + n \log T_2$

$\Rightarrow n = \frac{\log (V_2/V_1)}{\log (T_1/T_2)} = 0.2 \Rightarrow C = VT^n = 30(60)^{0.2} = 68$

$T = 30$ min, $V = ?$, $N = ?$ $d = 300$ mm $= 0.3$ m

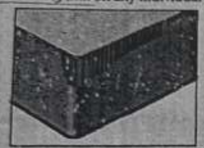
$VT^n = C \Rightarrow V(30)^{0.2} = 68 \Rightarrow V = 34.44$ m/min

$V = \pi D N \Rightarrow 34.44 = \pi \times 0.3 \times N \Rightarrow N = 36.54$ rpm

Tool life criteria ISO

(A certain width of flank wear (VB) is the most common criterion)

1. Uniform wear: 0.3 mm averaged over all past
2. Localized wear: 0.5 mm on any individual past



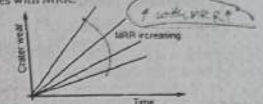
Crater wear

Due to diffusion of tool into chip material

- More common in ductile materials which produce continuous chip.
- Crater wear occurs on the rake face. at $v\uparrow$
- At very high speed crater wear predominates
- For crater wear temperature is main culprit and tool defuse into the chip material & tool temperature is maximum at some distance from the tool tip.

Crater wear Contd...

- Crater depth exhibits linear increase with time.
- It increases with MRR. ↑ with MRR↑



- Crater wear has little or no influence on cutting forces, work piece tolerance or surface finish.

Wear Mechanism

1. Abrasion wear flank wear
2. Adhesion wear micro weld
3. Diffusion wear High temp. crater wear.
4. Chemical or oxidation wear very-very high temp. + very high pr

[Diamond tool flooded coolant is used] bcoz it reacts with O_2

[Diamond is not used to cut Ferrous metal]

Why chipping off or fine cracks developed at the cutting edge

[the particles are removed from metal]

- Tool material is too brittle
- Weak design of tool, such as high positive rake angle ↑
- As a result of crack that is already in the tool
- Excessive static or shock loading of the tool.

$\sigma = \frac{F}{A}$

Stress concentration factor $K_t = 1 + \frac{2a}{b}$

Stress concentration is max at cutting edge

Notch Wear

Due to oxidation

- Notch wear on the trailing edge is to a great extent an oxidation wear mechanism occurring where the cutting edge leaves the machined workpiece material in the feed direction.
- But abrasion and adhesion wear in a combined effect can contribute to the formation of one or several notches.

List the important properties of cutting tool materials and explain why each is important.

- **Hardness at high temperatures** - this provides longer life of the cutting tool and allows higher cutting speeds.
- **Toughness** - to provide the structural strength needed to resist impacts and cutting forces
- **Wear resistance** - to prolong usage before replacement doesn't chemically react - another wear factor
- **Formable/manufacturable** - can be manufactured in a useful geometry

Tool Life Criteria

Tool life criteria can be defined as a predetermined numerical value of any type of tool deterioration which can be measured.

Some of the ways

- Actual cutting time to failure.
- Volume of metal removed.
- Number of parts produced.
- Cutting speed for a given time
- Length of work machined.

Taylor's Tool Life Equation

based on Flank Wear Tool life is a f^n of V only.

Causes

- Sliding of the tool along the machined surface
- Temperature rise

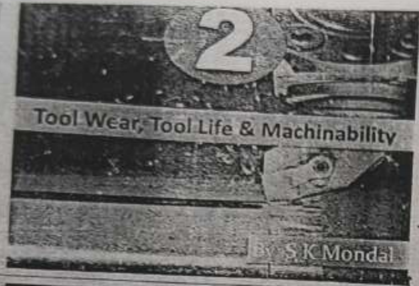
$$VT^n = C$$

Where, V = cutting speed (m/min)

T = Time (min)

n = exponent depends on tool material only

C = constant based on tool and work material and cutting condition.



Tool Failure

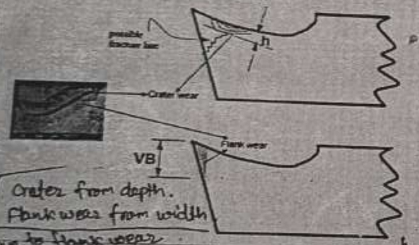
Tool failure is two types Tool life, based on slow death

- 1. **Slow-death:** The gradual or progressive wearing away of rake face (crater wear) or flank (flank wear) of the cutting tool or both.
- 2. **Sudden-death:** Failures leading to premature end of the tool
- The sudden-death type of tool failure is difficult to predict. Tool failure mechanisms include plastic deformation, brittle fracture, fatigue fracture or edge chipping. However it is difficult to predict which of these processes will dominate and when tool failure will occur.

Tool Wear

(a) Flank Wear
(b) Crater Wear
(c) Chipping off of the cutting edge

Tool Wear



Crater from depth.
Flank wear from width
due to flank wear

Flank Wear: (Wear land)

Reason

- Abrasion by hard particles and inclusions in the work piece.
- Shearing off the micro welds between tool and work material.
- Abrasion by fragments of built-up-edge ploughing against the clearance face of the tool.
- At low speed flank wear predominates.
- If MRR increased flank wear increased. $\boxed{\text{MRR} \uparrow \Rightarrow \text{flank wear} \uparrow}$

MRR ↑ ⇒ flank wear ↑

Flank Wear: (Wear land)

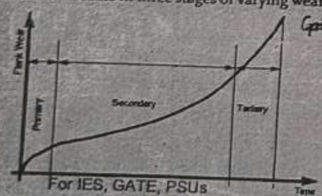
Effect

- Flank wear directly affect the component dimensions produced.
- Flank wear is usually the most common determinant of tool life.

Flank Wear: (Wear land)

Stages

- Flank Wear occurs in three stages of varying wear rates



Flank Wear: (Wear land)

Primary wear

The region where the sharp cutting edge is quickly broken down and a finite wear land is established.

Secondary wear

The region where the wear progresses at a uniform rate.

$\boxed{\text{Crater wear increases exponentially with temp.}}$

Flank Wear: (Wear land)

Tertiary wear

The region where wear progresses at a gradually increasing rate.

- In the tertiary region the wear of the cutting tool has become sensitive to increased tool temperature due to high wear land.
- Re-grinding is recommended before they enter this region.

Formula

$$V_o T_o^n = C$$

Optimum tool life for minimum cost

$$T_o = \left(T_c + \frac{C_t}{C_m}\right)\left(\frac{1-n}{n}\right) \quad \text{if } T_c, C_t \,\&\, C_m \text{ given} \qquad C_t = K_2$$

$$= \frac{C_t}{C_m}\left(\frac{1-n}{n}\right) \quad \text{if } C_t \,\&\, C_m \text{ given} \qquad C_m = K_1$$

Optimum tool life for Maximum Productivity (minimum production time)

$$T_o = T_c\left(\frac{1-n}{n}\right)$$

Units: T_c - min (Tool changing time)

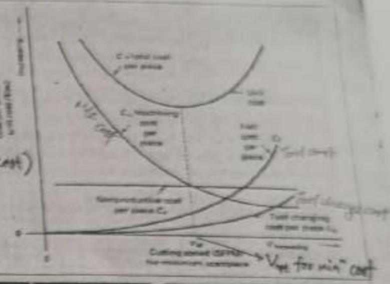
C_t - Rs./ servicing or replacement (Tooling cost)

C_m - Rs/min (Machining cost)

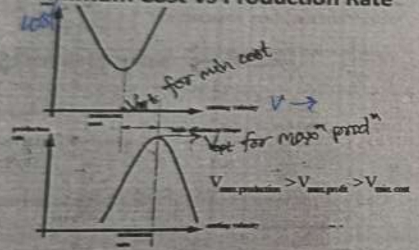
V - m/min (Cutting speed)

Tooling cost (C_t) = tool regrind cost (grinding time × machining cost) + tool depreciation per service/ replacement

Machining cost (C_m) = labour cost + over head cost per min



Minimum Cost Vs Production Rate



Machinability

Machinability-Definition

Machinability can be tentatively defined as 'ability of being machined' and more reasonably as 'ease of machining'.

Such ease of machining or machining characters of any tool-work pair is to be judged by:

- Tool wear or tool life
- Magnitude of the cutting forces
- Surface finish Copper & Brass excellent surface finish
- Magnitude of cutting temperature
- Chip forms. Discontinuous chip (cast iron)

Machinability ↑ if F↓, T↓, V_f↓, tool wear↓ Contd……:

- Machinability will be high when cutting forces, temperature, surfaces roughness and tool wear are less, tool life is long and chips are ideally uniform and short.
- The addition of sulphur, lead and tellurium to non-ferrous and steel improves machinability. ↑ m/c
- Sulphur is added to steel only if there is sufficient manganese in it. Sulphur forms manganese sulphide which exists as an isolated phase and act as internal lubrication and chip breaker.
- If insufficient manganese is there a low melting iron sulphide will formed around the austenite grain boundary. Such steel is very weak and brittle.

Manganese sulphide acts as solid lubricant & chip breaker.

Free Cutting steels

Low carbon steel

- Addition of lead in low carbon re-sulphurised steels and also in aluminium, copper and their alloys help reduce their τ_s. The dispersed lead particles act as discontinuity and solid lubricants and thus improve machinability by reducing friction, cutting forces and temperature, tool wear and BUE formation.
- It contains less than 0.35% lead by weight.
- A free cutting steel contains

C-0.07%, Si-0.03%, Mn-0.9%, P-0.04%, S-0.22%, Pb-0.15%

Machinability Index Or Machinability Rating

The machinability index K_M is defined by

$$K_M = V_{60}/V_{60R}$$

Where V_{60} is the cutting speed for the target material that ensures tool life of 60 min, V_{60R} is the same for the reference material.

If $K_M > 1$, the machinability of the target material is better that this of the reference material, and vice versa

Q. In certain machining oprⁿ with a cutting speed of 50 m/min, tool life of 45 min. was observed. When cutting speed was increased to 100 m/min, the tool life decreased to 10 min. Estimate the cutting speed for maxᵐ productivity if tool change time is 2 min.

Ans. $T_0 = T_c\left(\frac{1-n}{n}\right)$

$VT^n = C \Rightarrow V_1 T_1^n = V_2 T_2^n$

$50(45)^n = 100(10)^n \Rightarrow n = 0.46$

$C = 288.04$

$$V = \frac{C}{\left[T_c\left(\frac{1-n}{n}\right)\right]^n} = \frac{288.04}{2\times\left(\frac{1-0.46}{0.46}\right)^{0.46}}$$

$= 194.5$ m/min.

Ques. Determine to optimum cutting speed for an oprⁿ on a lathe using the following information:

(i) Tool change time : 3 min | Machine running cost : Rs 0.50 per min

(ii) Tool regrind time : 3 min | Depreciation of tool regrind = Rs. 5.0 ⇒ C_m or K_2

The constants in the tool life eqⁿ are 60 and 0.2

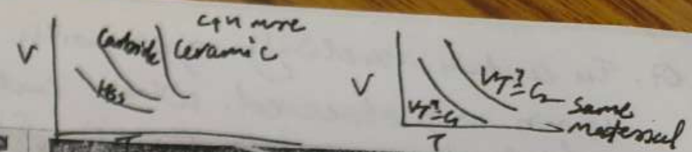
Ans. Direct running cost $= C_t$ or K_1 = Regrind time × running cost/unit time

$= 3\text{ min} \times 0.5/\text{min} \Rightarrow$ Rs 1.5

$$T_0 = \left(\frac{1-n}{n}\right)\left(T_c + \frac{K_2}{K_1}\right) \Rightarrow V_{opt} \text{ for min cost} = \frac{C}{\left[\left(\frac{1-n}{n}\right)\left(T_c + \frac{K_2}{K_1}\right)\right]^n} = \frac{60}{\left[\left(\frac{1-0.2}{0.2}\right)\left(3 + \frac{5}{1.5}\right)\right]^{0.2}}$$

$V_{opt} = 31.43$ m/min

$$\text{optᵐ cutting speed for maxᵐ prodⁿ} = \frac{C}{\left[T_c\left(\frac{1-n}{n}\right)\right]^n} = \frac{60}{\left[\left(\frac{1-0.2}{0.2}\right)\times 3\right]^{0.2}} = 36.5 \text{ m/min}$$



Values of Exponent 'n'

n = 0.08 to 0.2 for HSS tool
= 0.1 to 0.15 for Cast Alloys
= 0.2 to 0.4 for carbide tool
[IAS-1999; IES-2006]
= 0.5 to 0.7 for ceramic tool
[NTPC-2003]

Extended or Modified Taylor's equation

$$VT^n f^a d^b = C$$

Where: d = depth of cut
f = feed rate

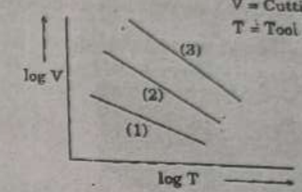
or $T = \dfrac{C^{1/n}}{V^{1/n}.f^{1/n_1}.d^{1/n_2}}$

$$\frac{1}{n} > \frac{1}{n_1} > \frac{1}{n_2}$$ effects $V > f > d$

i.e Cutting speed has the greater effect followed by feed and depth of cut respectively.

Tool Life Curve

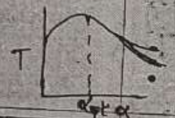
V = Cutting speed
T = Tool life



1. HSS 2. Carbide 3. Ceramic

Cutting speed used for different tool materials

HSS (min) 30 m/min < Cast alloy < Carbide < Cemented carbide 150 m/min < Cermets < Ceramics or sintered oxide (max) 600 m/min



Effect of Rake angle on tool life

- If rake angle is large, smaller will be cutting angle and larger will be the shear angle. This will reduce force and power of cut i.e. ↑ tool life.
- But increasing the rake angle reduces the mass of metal behind the cutting edge resulting in poor transfer of heat i.e. ↓ tool life.
- Therefore optimum value of $\boxed{\alpha = 14^0}$

$\alpha\uparrow \Rightarrow C_{\alpha}\downarrow \Rightarrow \phi\uparrow \Rightarrow$ Force↓ ⇒ Tool Life↑

Effect of Clearance angle on tool life

If clearance angle increased it reduces flank wear but weaken the cutting edge, so best compromise is 8° for HSS and 5° for carbide tool. 8° HSS 5° Carbide Tool

Effect of work piece on tool life

- With hard micro-constituents in the matrix gives poor tool life.
- With larger grain size tool life is better.

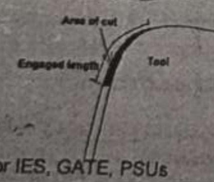
Grain size↑ ⇒ Tool Life↑

Chip Equivalent

$$\text{Chip Equivalent}(q) = \frac{\text{Engaged cutting edge length}}{\text{Plan area of cut}}$$

- It is used for controlling the tool temperature.

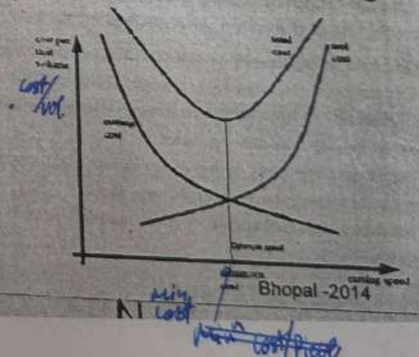
q more Tool Life more



$C_s\uparrow \Rightarrow q\uparrow \Rightarrow T\uparrow$ tool life
$r\uparrow \Rightarrow q\uparrow$

- The SCEA alters the length of the engaged cutting edge without affecting the area of cut. As a result, the chip equivalent changed. When the SCEA is increased, the chip equivalent is increased, without significantly changing the cutting forces.
- Increase in nose radius also increases the value of the chip equivalent and improve tool life.

Economics of metal cutting



Role of microstructure on Machinability

Coarse microstructure leads to lesser value of τ_s.

Therefore, τ_s can be desirably reduced by Annealing ⇒ $\tau_s\downarrow$

- Proper heat treatment like annealing of steels
- Controlled addition of materials like sulphur (S), lead (Pb), Tellerium etc leading to free cutting of soft ductile metals and alloys.
- Brittle materials are relatively more machinable.

Effects of tool rake angle(s) on machinability

$\alpha\uparrow \Rightarrow m/c\uparrow$

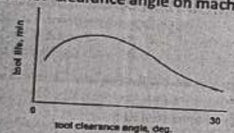
- As Rake angle increases machinability increases.
- But too much increase in rake weakens the cutting edge.

Effects of Cutting Edge angle(s) on machinability

- The variation in the cutting edge angles does not affect cutting force or specific energy requirement for cutting.
- Increase in SCEA and reduction in ECEA improves surface finish sizeably in continuous chip formation hence Machinability.

$C_s\uparrow + C_e\downarrow$ m/c↑

Effects of clearance angle on machinability



Inadequate clearance angle reduces tool life and surface finish by tool – work rubbing, and again too large clearance reduces the tool strength and tool life hence machinability.

Effects of Nose Radius on machinability

Proper tool nose radiusing improves machinability to some extent through

- increase in tool life by increasing mechanical strength and reducing temperature at the tool tip
- reduction of surface roughness, h_{max}

$R\uparrow \Rightarrow h\downarrow \Rightarrow m/c\uparrow$

$$\downarrow h_{max} = \frac{f^2}{8R\uparrow}$$

Surface Roughness

IMP (Turning ope...)

- **Ideal Surface (Zero nose radius)**

Peak to valley roughness (h) = $\dfrac{f}{\tan SCEA + \cot ECEA}$

and $(R_a) = \dfrac{h}{4} = \dfrac{f}{4(\tan SCEA + \cot ECEA)}$

- **Practical Surface (with nose radius = R)**

$h = \dfrac{f^2}{8R}$ and $R_a = \dfrac{f^2}{18\sqrt{3}R}$ → Roughness average

Change in feed (f) is more important than a change in nose radius (R) and Depth of cut has no effect on surface roughness.

Cutting fluid

- The cutting fluid acts primarily as a coolant and secondly as a lubricant, reducing the friction effects at the tool-chip interface and the work-blank regions.
- **Cast Iron:** Machined dry or compressed air, Soluble oil for high speed machining and grinding
- **Brass:** Machined dry or straight mineral oil with or without EPA.
- **Aluminium:** Machined dry or kerosene oil mixed with mineral oil or soluble oil
- **Stainless steel and Heat resistant alloy:** High performance soluble oil or neat oil with high concentration with chlorinated EP additive.

Extreme pressure

primary → coolant
→ lubricant

IAS -2009 Main

- What are extreme-pressure lubricants? [3 – marks]

Where high pressures and rubbing action are encountered, hydrodynamic lubrication cannot be maintained; so Extreme Pressure (EP) additives must be added to the lubricant. EP lubrication is provided by a number of chemical components such as boron, phosphorus, sulfur, chlorine, or combination of these. The compounds are activated by the higher temperature resulting from extreme pressure. As the temperature rises, EP molecules become reactive and release derivatives such as iron chloride or iron sulfide and forms a solid protective coating.

lubricant (Hydrodynamic lubrication)